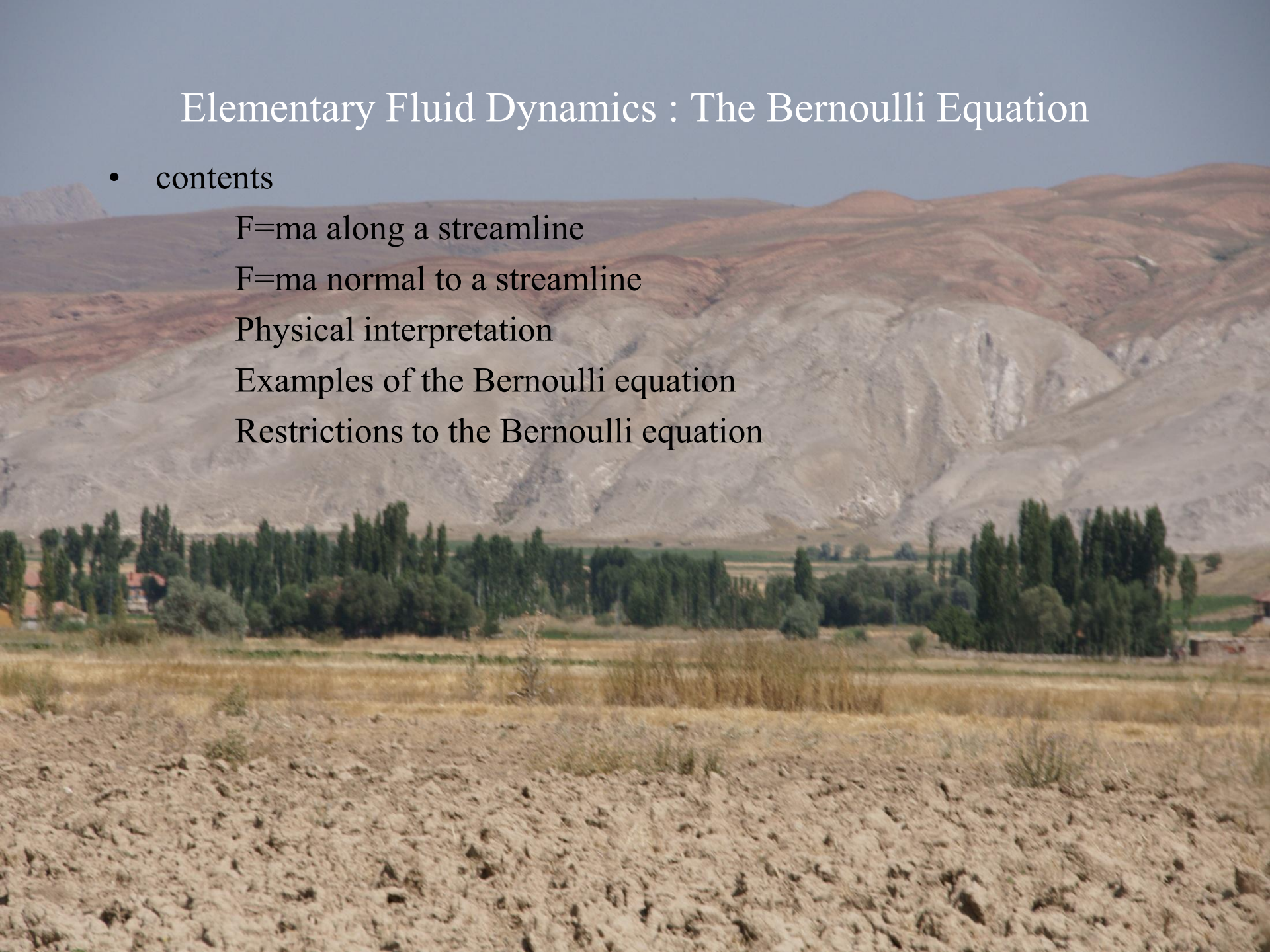
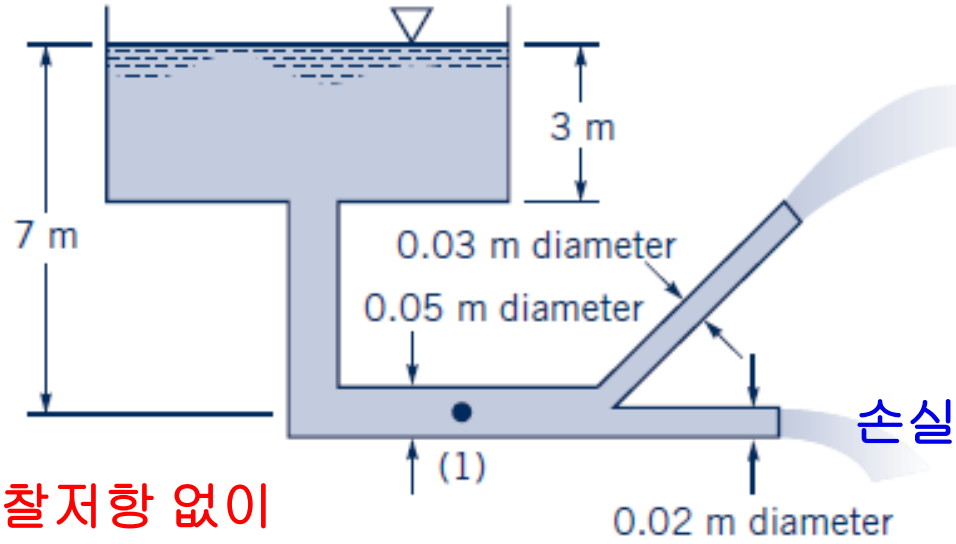
Elementary Fluid Dynamics : The Bernoulli Equation

- contents
  - F=ma along a streamline
  - F=ma normal to a streamline
  - Physical interpretation
  - Examples of the Bernoulli equation
  - Restrictions to the Bernoulli equation



제4장에서 공부할 주제들 Subject to study in chapter 4



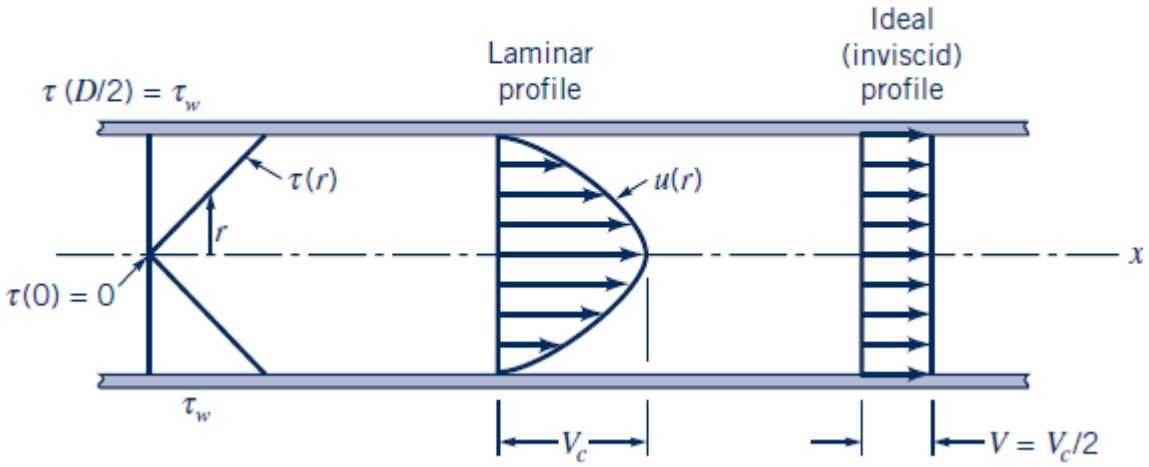
$Q_{w/o\ \ friction} ? Q_{with\ \ friction}$

손실 Losses : chapter 7 & 8에서 공부

마찰저항 없이
Without friction

4.1 베르누이 방정식의 한계 Limitations of the Bernoulli equation

- 점성 대 비점성 **Viscous vs. ideal flows**



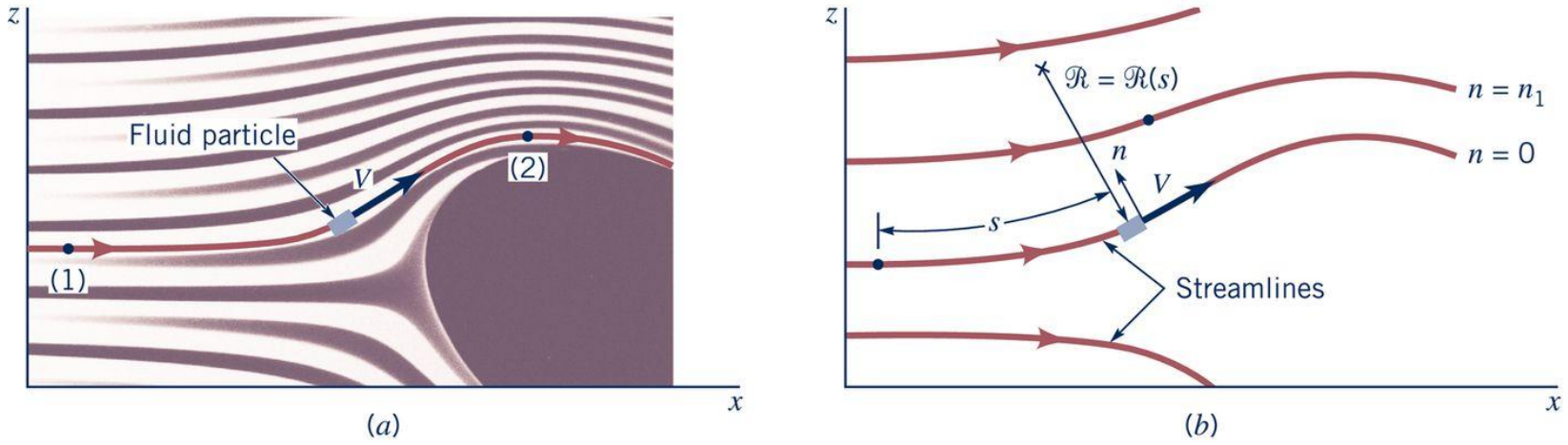
- 점성력 무시 **Neglect viscous force**

  => 이상적 유동 ideal flow

  => 에너지 손실 없음 no energy loss

4.2 F=ma 유선을 따라서 along a streamline

- 유선 좌표계 **Streamline coordinates**



- 유선을 따른 가속도 **Acceleration along a streamline**

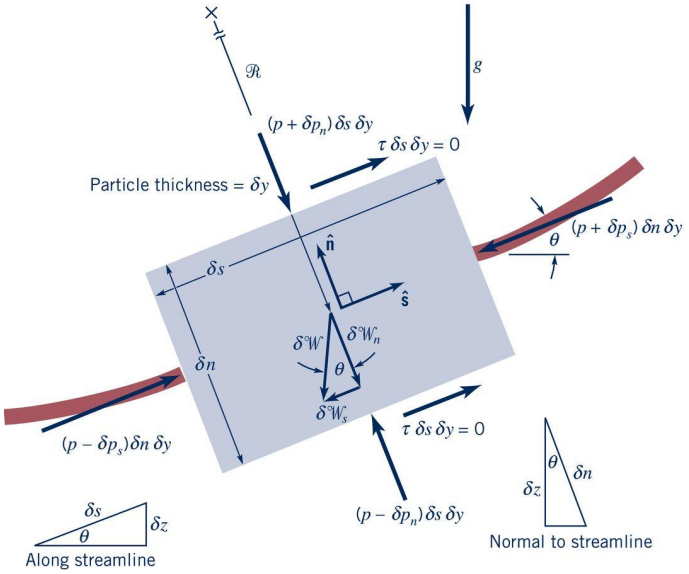
$$a_s = \frac{dV}{dt} \Rightarrow \frac{DV}{Dt} = \frac{\partial V}{\partial t} + \frac{\partial V}{\partial s}\frac{ds}{dt} + \frac{\partial V}{\partial n}\frac{dn}{dt}$$

$$\therefore a_s = \frac{dV}{dt} = \frac{\partial V}{\partial s}\frac{ds}{dt} = V\frac{\partial V}{\partial s} : steady\ flow\ along\ a\ streamline$$

유선을 따라가는 정상 흐름

4.2 F=ma 유선을 따라서 along a streamline

- 뉴톤의 운동법칙 **Newton's law of motion**



$$\sum \delta F_s = \delta m \cdot a_s = (\rho \cdot \delta\forall) \cdot V\frac{\partial V}{\partial s} : steady\ flow$$

$$\sum \delta F_s = \sum [(\delta F_s)_{body} + (\delta F_s)_{surface}]$$

$$\sum (\delta F_s)_{body} = -\delta W \cdot sin\theta = -\gamma \cdot \delta\forall \cdot sin\theta$$

$$\sum (\delta F_s)_{surface} = (p - \delta p_s)\delta n \cdot \delta y - (p + \delta p_s)\delta n \cdot \delta y = -2\delta p_s \cdot \delta n \cdot \delta y$$

$$since\ \delta p_s = \frac{\partial p}{\partial s}\frac{\delta s}{2} \Rightarrow \sum (\delta F_s)_{body} = -\frac{\partial p}{\partial s}\delta s \cdot \delta n \cdot \delta y = -\frac{\partial p}{\partial s}\delta\forall$$

$$\therefore \sum \delta F_s = -\gamma \cdot \delta\forall \cdot sin\theta - \frac{\partial p}{\partial s}\delta\forall = (\rho \cdot \delta\forall) \cdot V\frac{\partial V}{\partial s}$$

$$\therefore -\gamma \cdot sin\theta - \frac{\partial p}{\partial s} = \rho V\frac{\partial V}{\partial s}$$

$$\therefore -\gamma\frac{dz}{ds} - \frac{\partial p}{\partial s} = \rho V\frac{\partial V}{\partial s}$$

4.2 F=ma 유선을 따라서 along a streamline

- 유선을 따라서 **Along a streamline**

$$-\gamma\frac{dz}{ds}-\frac{\partial p}{\partial s}=\rho V\frac{\partial V}{\partial s}\Rightarrow -\gamma\frac{dz}{ds}-\frac{dp}{ds}=\rho V\frac{dV}{ds}$$

$$\therefore V\frac{dV}{ds}+\frac{1}{\rho}\frac{dp}{ds}+g\frac{dz}{ds}=0$$

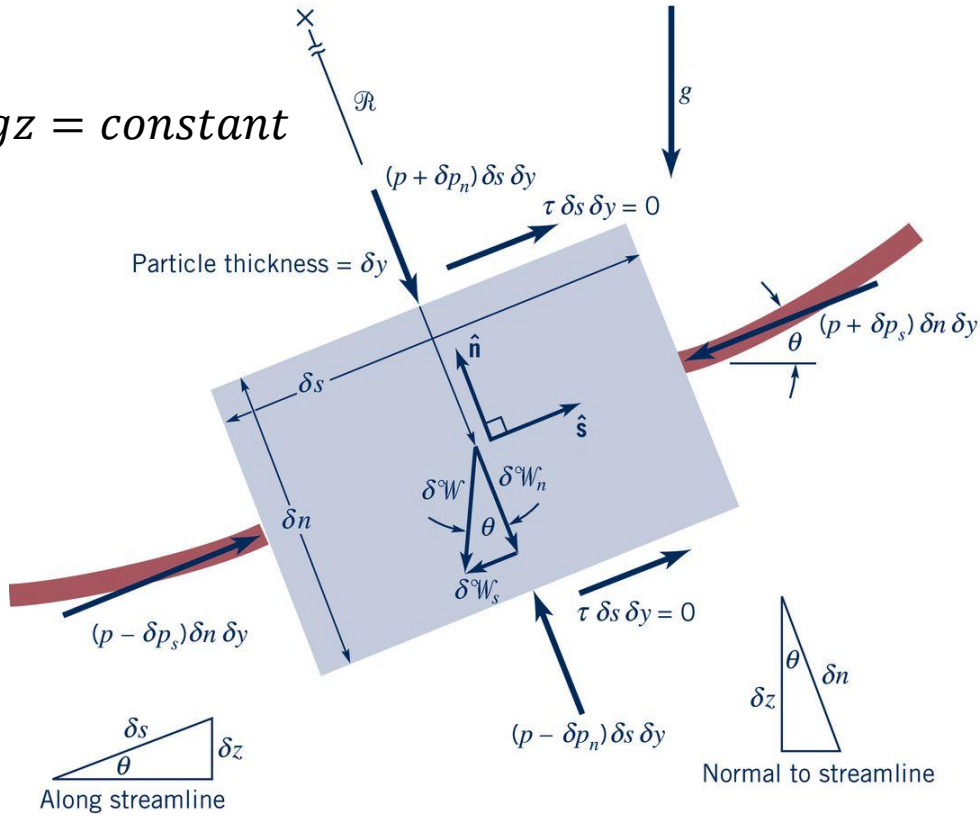
Integrating

$$\int\left(V\frac{dV}{ds}+\frac{1}{\rho}\frac{dp}{ds}+g\frac{dz}{ds}\right)ds=\frac{V^2}{2}+\int\frac{dp}{\rho}+gz=constant$$

- **If incompressible** 비압축성이면

$$\frac{V^2}{2}+\frac{p}{\rho}+gz=constant:incompressible$$

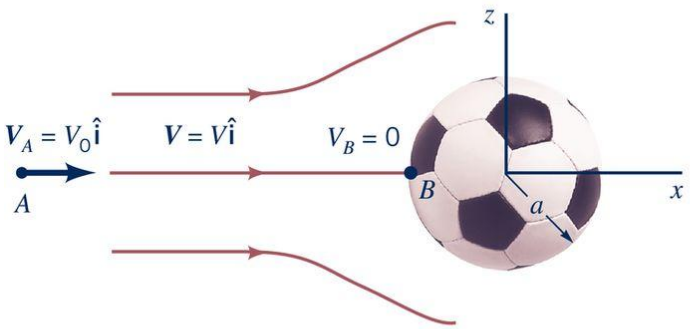
$$or\quad \frac{V_1^2}{2}+\frac{p_1}{\rho}+gz_1=\frac{V_2^2}{2}+\frac{p_2}{\rho}+gz_2$$



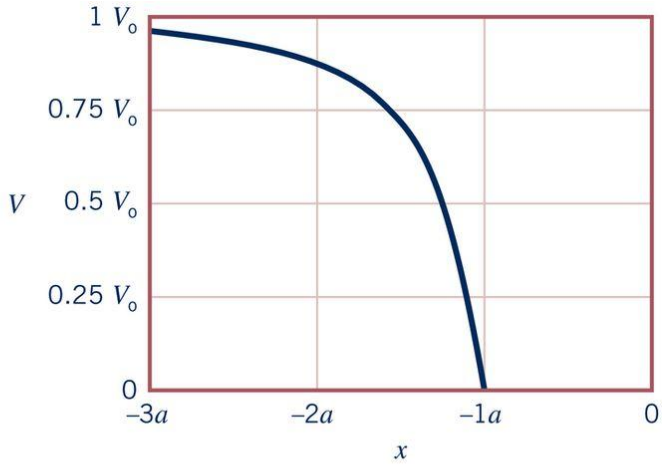
4.2 F=ma 유선을 따라서 along a streamline

Ex4.1) From a more advanced theory of flow past a sphere, the fluid along the streamline A-B is $V=V_o(1+a^3/x^3)$. Determine the pressure variation along the streamline from point A in front of the sphere to point B on the sphere.

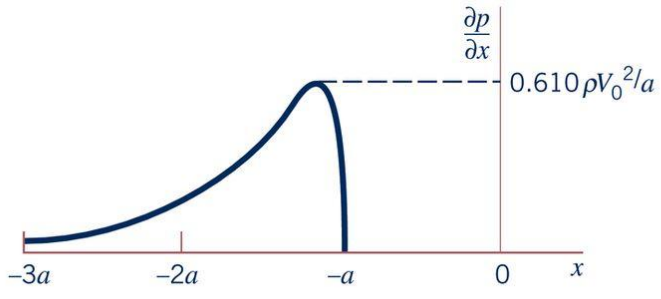
Ex 4.1) 구 주위 유동에 대한 보다 고등 이론으로부터, 유선 A-B 을 따른 유속은 $V=V_o(1+a^3/x^3)$이다. 구 앞 쪽의 점 A에서 구 표면의 점 B에 이르는 유선을 따른 압력변화를 구하라.



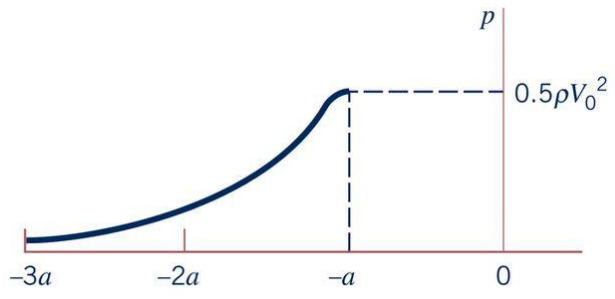
(a)



(b)



(c)



(d)

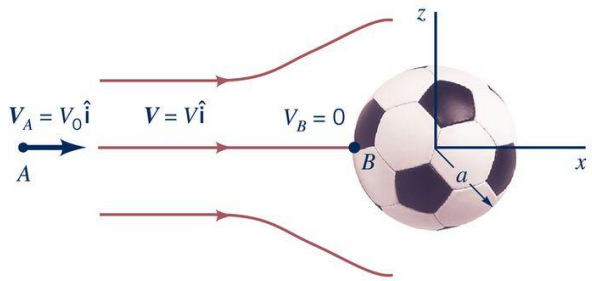
4.2 F=ma 유선을 따라서 along a streamline

- 베르누이 방정식 **Bernoulli equation**

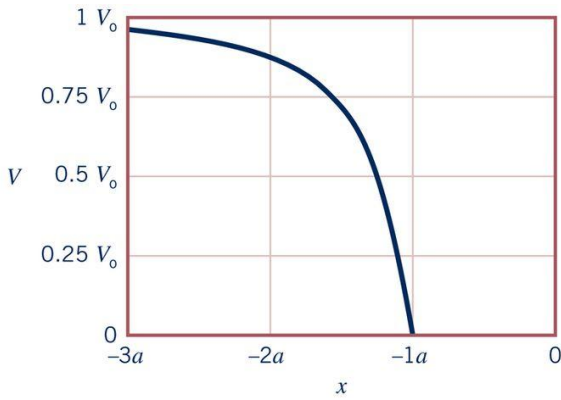
$$\frac{dp}{ds} = -\rho V\frac{dV}{ds} - \gamma\frac{dz}{ds} = -\rho V\frac{dV}{ds} = -\rho V\frac{dV}{dx}$$

$$\therefore \frac{dp}{ds} = \frac{dp}{dx} = -\rho V_0\left(1+\frac{a^3}{x^3}\right)\frac{d[V_0(1+\frac{a^3}{x^3})]}{dx} = -3\rho V_0^2\left(1+\frac{a^3}{x^3}\right)\frac{a^3}{x^4} = -3\rho V_0^2(\frac{a^3}{x^4}+\frac{a^3}{x^7})$$

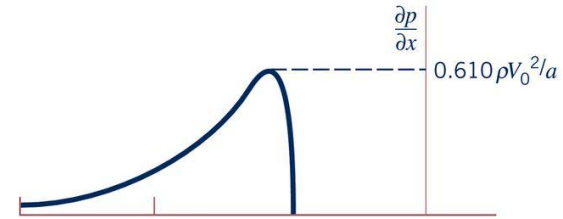
$$\therefore p = -3\rho V_0^2(\frac{a^3}{3x^3}+\frac{a^6}{6x^6}) + p_0$$



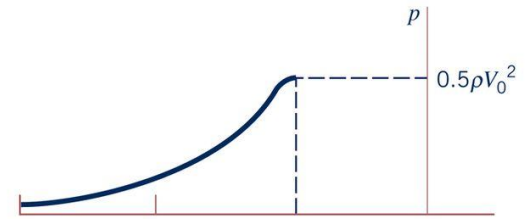
(a)



(b)



(c)

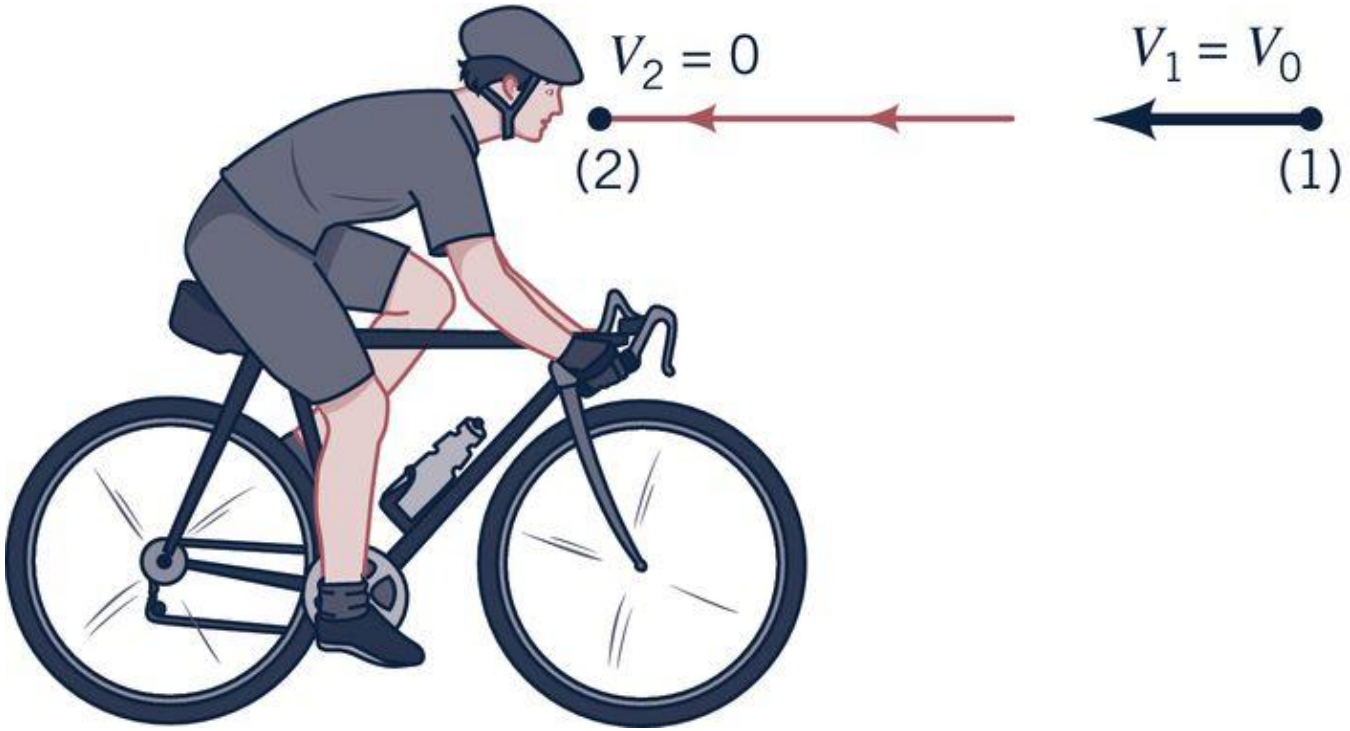


(d)

4.2 F=ma 유선을 따라서 along a streamline

Ex3.2) Consider the flow of air around a bicyclist moving through still air with velocity V_o. Determine the difference in the pressure between points (1) and (2).

Ex 3.2) 정지된 공기 중에서 속도 V_o로 운동하는 자전거 타는 사람 주위의 공기 유동을 고려해보자. 점 (1)과 (2) 사이의 압력 차이를 구하라.



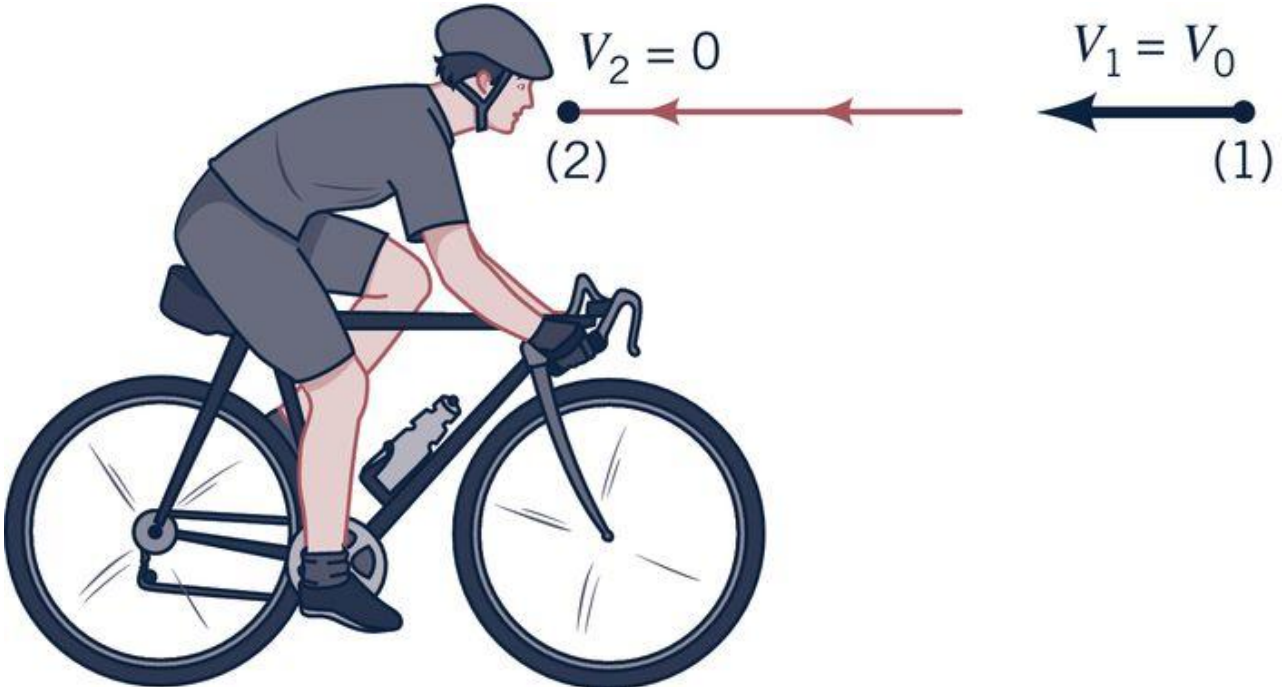
4.2 F=ma 유선을 따라서 along a streamline

- 베르누이 방정식 **Bernoulli equation**

$$p + \frac{\rho V^2}{2} + \gamma z = constant$$

$$\therefore p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_2 + \frac{\rho V_2^2}{2} + \gamma z_2$$

$$\therefore p_2 - p_1 = \frac{\rho V_1^2}{2} - \frac{\rho V_2^2}{2} + \gamma(z_1 - z_2) = \frac{\rho V_0^2}{2}$$



4.3 F=ma 유선에 수직한 방향으로 normal to a streamline

- **Newton's law of motion**

$$\sum \delta F_n = \delta m \cdot a_n = (\rho \cdot \delta \forall) \cdot a_n$$

- 가속도 **Acceleration**

$$V_n = 0, \qquad \therefore a_n = \frac{V^2}{R}$$

- Forces

$referecne)$

$$\vec{a} = a_s\vec{i_s} + a_n\vec{i_n} = \frac{d(V\vec{i_s})}{dt} = \frac{d(V\vec{i_s})}{ds}\frac{ds}{dt}$$

$$= \left(\frac{dV}{ds}\vec{i_s} + V\frac{d\vec{i_s}}{ds}\right)\frac{ds}{dt}$$

$$\frac{d\vec{i_s}}{ds} = \frac{d(cos\theta\vec{i} + sin\theta\vec{j})}{Rd\theta} = \frac{1}{R}(-sin\theta\vec{i} + cos\theta\vec{j}) = \frac{\vec{i_n}}{R}$$

$$\sum \delta F_n = \sum [(\delta F_n)_{body} + (\delta F_n)_{surface}]$$

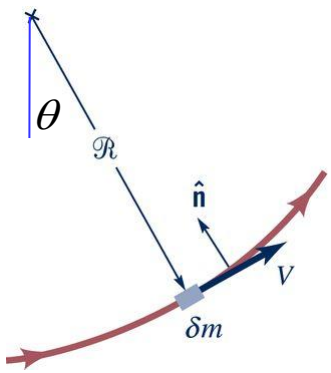
$$\therefore \vec{a} = a_s\vec{i_s} + a_n\vec{i_n} = V\frac{dV}{ds}\vec{i_s} + \frac{V^2}{R}\vec{i_n}$$

$$\sum (\delta F_n)_{body} = -\delta W \cdot cos\theta = -\gamma \cdot \delta \forall \cdot cos\theta$$

$$\sum (\delta F_n)_{surface} = (p - \delta p_n)\delta s \cdot \delta y - (p + \delta p_n)\delta s \cdot \delta y = -2\delta p_n \cdot \delta s \cdot \delta y$$

$$since\ \delta p_n = \frac{\partial p}{\partial n}\frac{\delta n}{2} \Rightarrow \sum (\delta F_n)_{body} = -\frac{\partial p}{\partial n}\delta s \cdot \delta y \cdot \delta n = -\frac{\partial p}{\partial n}\delta \forall$$

$$\therefore \sum \delta F_n = -\gamma \cdot \delta \forall \cdot cos\theta - \frac{\partial p}{\partial n}\delta \forall$$



4.3 F=ma 유선에 수직한 방향으로 normal to a streamline

- **F=ma** 유선에 수직한 방향으로 **normal to a streamline**

$$\therefore \sum \delta F_n = -\gamma \cdot \delta\forall \cdot cos\theta - \frac{\partial p}{\partial n}\delta\forall = (\rho \cdot \delta\forall) \cdot a_n$$

$$\Rightarrow -\gamma \cdot \delta\forall \cdot cos\theta - \frac{\partial p}{\partial n}\delta\forall = (\rho \cdot \delta\forall) \cdot \frac{V^2}{R}$$

$$\therefore -\gamma\frac{dz}{dn} - \frac{\partial p}{\partial n} = \rho\frac{V^2}{R} \Rightarrow \therefore -g\frac{dz}{dn} - \frac{1}{\rho}\frac{\partial p}{\partial n} = \frac{V^2}{R}$$

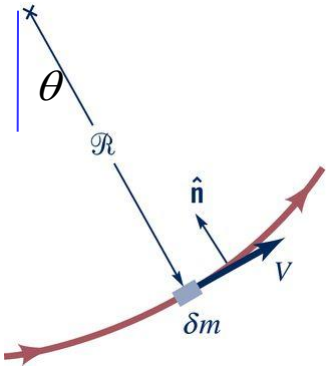
- 적분하면 **Integrating**

유선에 수직한 방향으로

$$\int \frac{dp}{\rho} + \int \frac{V^2}{R} dn + gz = constant: across\ a\ streamline$$

- 비압축성이라면 **If incompressible**

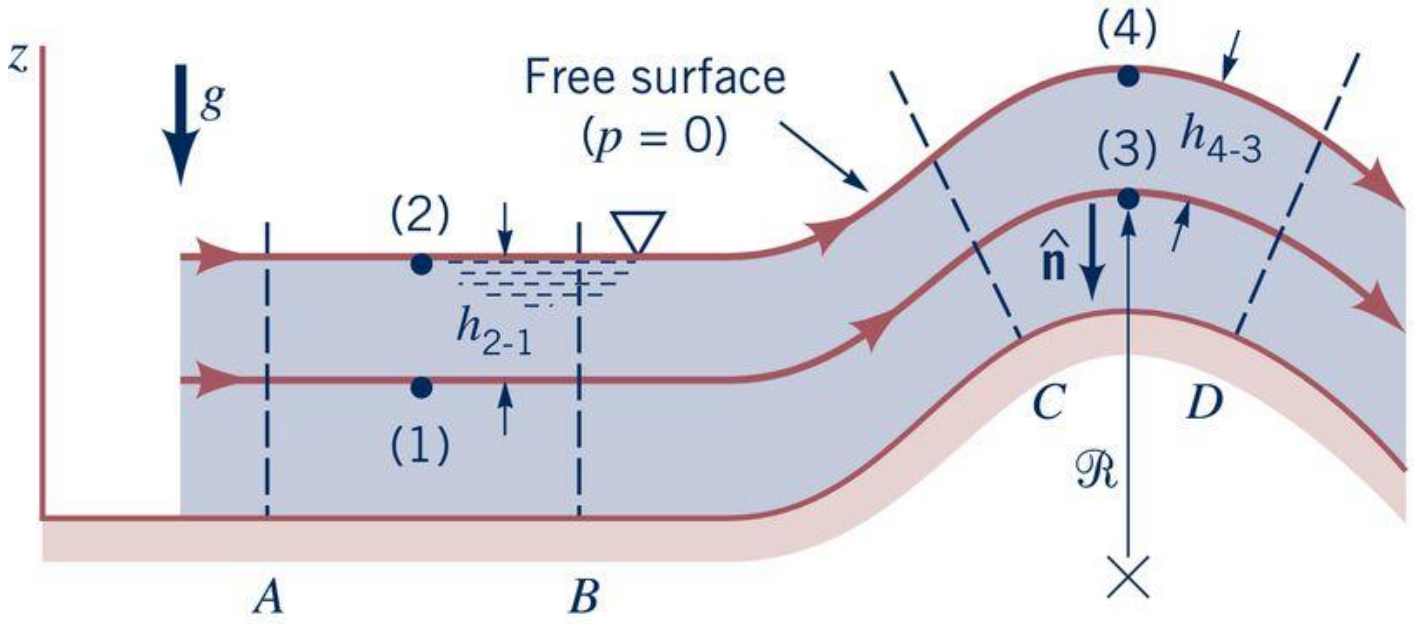
$$\frac{p_2 - p_1}{\rho} + \int_1^2 \frac{V^2}{R} dn + g(z_2 - z_1) = 0$$



4.4 물리적 해석 Physical interpretation

Ex4.5) Water flows in a curved, undulating waterslide as shown in the figure. As an approximation to this flow, consider the inviscid, incompressible, steady flow. From section A to section B the streamlines are straight, while from C to D they follow circular paths. Determine the pressure variation between points (1) and (2) and points (3) and (4).

Ex 4.5) 바닥이 굴곡지고 굽은 물 미끄럼틀에 물이 흐른다. 이 유동을 비점성, 비압축성, 정상흐름이라고 가정하자. 유선이 단면 A에서 B까지는 직선이고, C에서 D까지는 원형이다. 점 (1)과 (2) 사이 그리고 (3)과 (4) 사이의 압력변화를 구하라.



4.4 물리적 해석 Physical interpretation

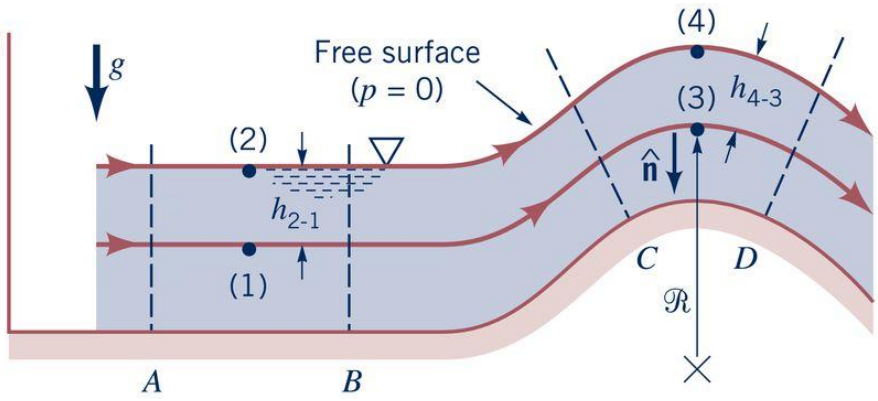
- 직선 유선을 가로질러 **Across the straight streamline**

$$\frac{p_2 - p_1}{\rho} + \int_1^2 \frac{V^2}{R} dn + g(z_2 - z_1) = 0$$

$$since R \to \infty$$

$$\therefore \frac{p_2 - p_1}{\rho} + g(z_2 - z_1) = 0$$

$$\therefore p_1 = p_2 + \rho g(z_2 - z_1) = p_2 + \gamma h_{2-1}$$



- 원형 유선을 가로질러 **Across the circular streamline (3)**

$$\frac{p_4 - p_3}{\rho} + \int_3^4 \frac{V^2}{R} dn + g(z_4 - z_3) = 0$$

$$\therefore p_3 = p_4 + \rho g(z_4 - z_3) + \int_3^4 \frac{V^2}{R} dn$$

$$since\ dn = -dz$$

$$p_3 = p_4 + \rho g(z_4 - z_3) - \int_3^4 \frac{V^2}{R} dz < \gamma(z_4 - z_3)$$

to accelerate the fluid along the curved path

4.5 정압, 정체압, 동압 및 전압
Static, Stagnation, Dynamic, and Total Pressure

- 베르누이 방정식 **Bernoulli equation**

$$p + \frac{\rho V^2}{2} + \gamma z = constant$$

- 정압 **Static,** 정수력압 **hydrostatic pressure, and** 동압 **dynamic pressures**

$p = static\ pressure,$ 정압 $\quad \gamma z = hydrostatic\ pressure,$ 정수력압 $\quad \frac{\rho V^2}{2} = dynamic\ pressure$ 동압

- 전압 **Total pressure**

$$p_{total} = p + \frac{\rho V^2}{2} + \gamma z$$

- 정체압 **Stagnation pressure**

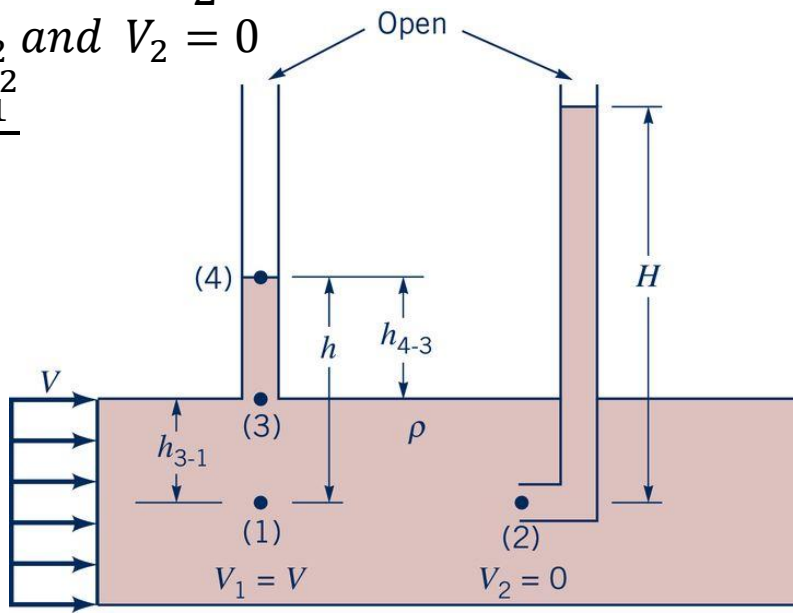
$$p_{stagnation} = p + \frac{\rho V^2}{2}$$

Bernoulli equation along the streamline (1)*and* (2)

$$p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_2 + \frac{\rho V_2^2}{2} + \gamma z_2$$

$since\ z_1 = z_2\ and\ V_2 = 0$

$$\therefore p_2 = p_1 + \frac{\rho V_1^2}{2}$$



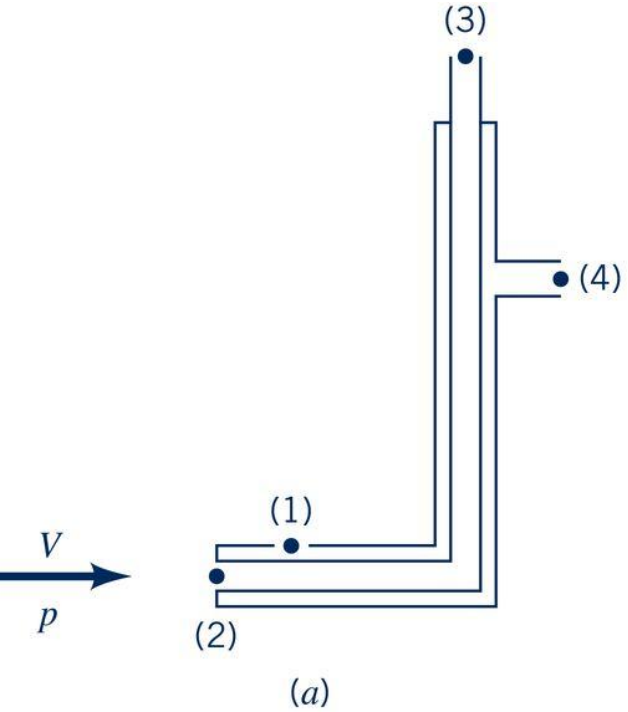
4.5 정압, 정체압, 동압 및 전압
Static, Stagnation, Dynamic, and Total Pressure

- 피토정압관 **Pitot-static tube**

$$p_3 = p + \frac{\rho V^2}{2} \quad and \quad p_4 = P_1 = p$$

$$\therefore p_3 - p_4 = \frac{\rho V^2}{2}$$

$$\therefore V = \sqrt{\frac{2(p_3 - p_4)}{\rho}}$$

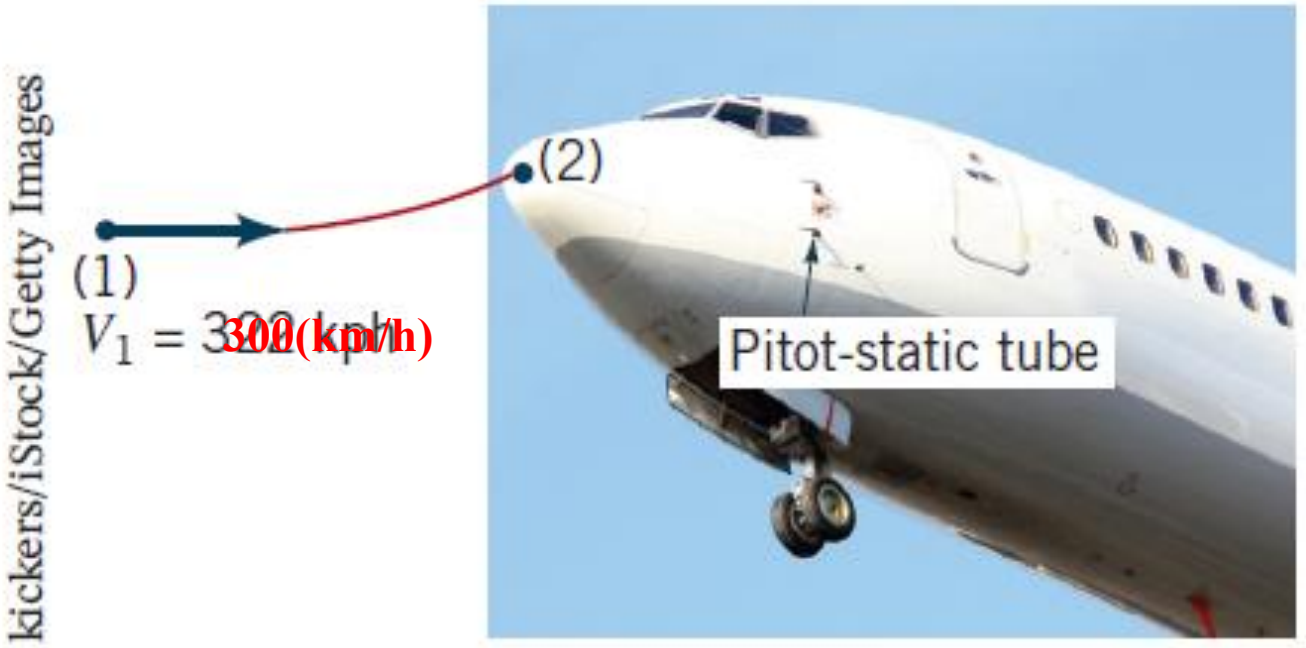




4.5 정압, 정체압, 동압 및 전압
Static, Stagnation, Dynamic, and Total Pressure

Ex4.6) An airplane files 300(km/h) at an altitude of 3000(m) in a standard atmosphere as shown below. Determine the pressure at point (1) far ahead of the air plane, the pressure at the stagnation point on the nose of the airplane, point (2), and the pressure difference indicated by a pitot-static attached to the fuselage.

Ex 4.6) 표준대기 고도 3000(m)에서 300(km/h)로 비행기가 날고 있다. 비행기 앞 점 (1)과 비행기의 코 부근 정체점 (2)에서의 압력과 동체에 부착된 피토정압관에서의 압력차이를 구하라.



4.5 정압, 정체압, 동압 및 전압
Static, Stagnation, Dynamic, and Total Pressure

- **(a) 점(1)에서의 압력 pressure at point (1)**

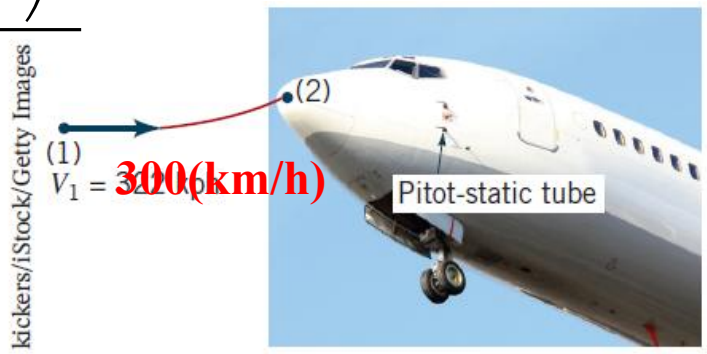
$$p_1 = 70.12(kPa) = from\ the\ standard\ atmospheric\ data$$

- **(b) 정체점에서의 압력 pressure at the stagnation point**

$$p_2 = p_1 + \frac{\rho V_1^2}{2} = 70.12(kPa) + \frac{0.9093\left(\frac{kg}{m^3}\right)\left(\frac{300{,}000(\frac{km}{h})}{3600(\frac{h}{s})}\right)^2}{2} = 73.252(kPa)$$

- **(c) 피토정압관 압력 차이 pressure difference indicated by a Pitot-static tube**

$$p_2 - p_1 = \frac{\rho V_1^2}{2} = \frac{0.9093\left(\frac{kg}{m^3}\right)\left(\frac{300{,}000\left(\frac{km}{h}\right)}{3600\left(\frac{h}{s}\right)}\right)^2}{2} = 3.132(kPa)$$



kickers/iStock/Getty Images

4.6 베르누이 방정식 응용 Use of the Bernoulli Equation

4.6.1 자유제트 Free Jets

- 점 (1)과 (2) 사이의 베르누이 방정식 Bernoulli equation between (1) and (2)

$$p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_2 + \frac{\rho V_2^2}{2} + \gamma z_2$$

$$p_1 = p_2 \approx p_{atm}, \qquad z_1 = h, \qquad z_2 = 0 \ \ and \ \ V_1 \ll V_2 = V$$

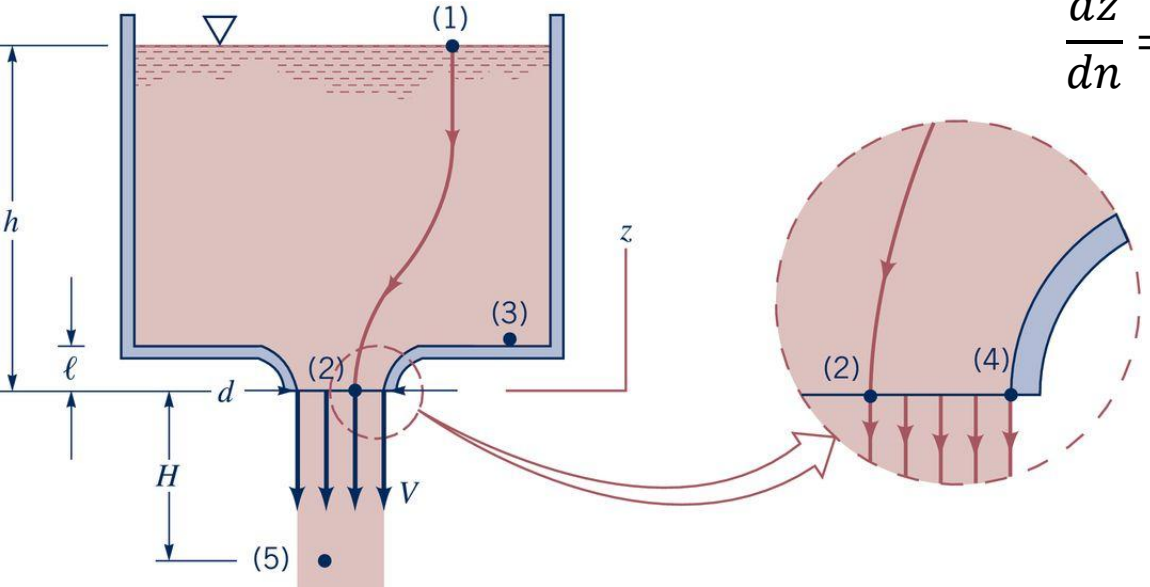
$$\therefore \gamma h = \frac{\rho V^2}{2} \Rightarrow V = \sqrt{2gh}$$

유선에 수직한 방향

$$ref)\ normal\ to\ a\ streamline$$

$$-g\frac{dz}{dn} - \frac{1}{\rho}\frac{\partial p}{\partial n} = \frac{V^2}{R}$$

$$\frac{dz}{dn} = 0\ and\ R \to \infty \therefore \frac{\partial p}{\partial n} = 0 \Rightarrow p_4 = p_2$$

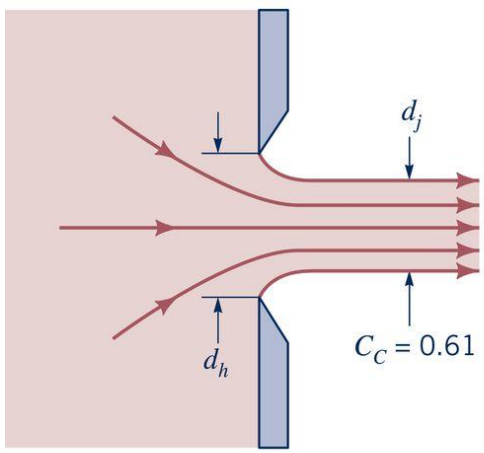


4.6 베르누이 방정식 응용 Use of the Bernoulli Equation

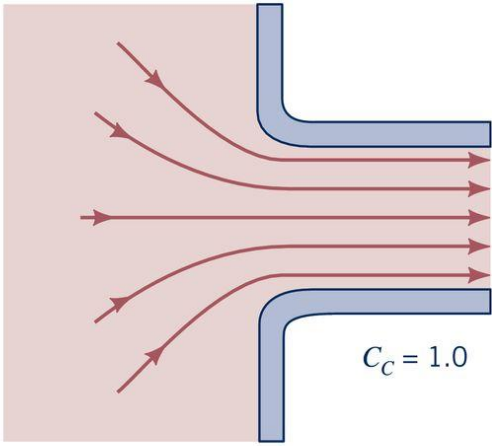
4.6.1 자유제트 Free Jets

- 수축계수 Contraction coefficient

$$C_c = \frac{A_j}{A_h}$$

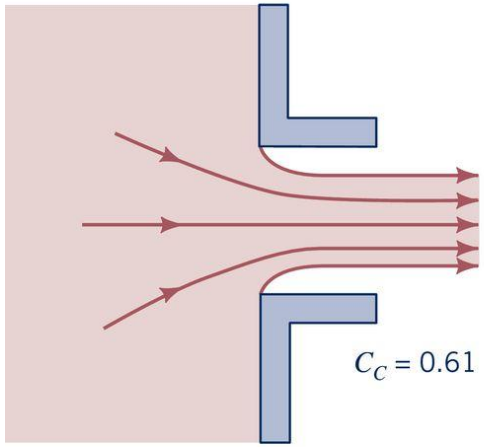


(a) Knife edge

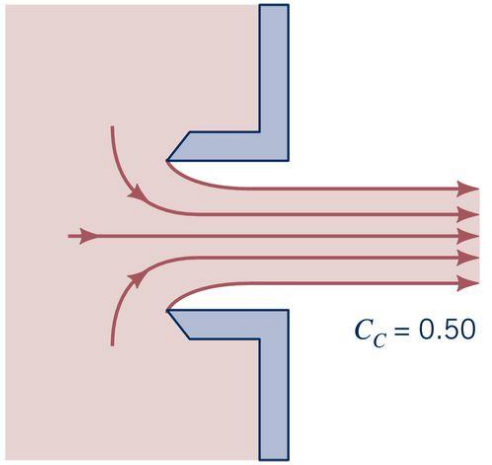


(b) Well rounded

$C_C = A_j/A_h = (d_j/d_h)^2$



(c) Sharp edge



(d) Re-entrant

4.6 베르누이 방정식 응용 Use of the Bernoulli Equation

4.6.2 갇힌 유동 Confined Flows

- 질량유량 **Mass flow rate**

$$\dot{m} = \rho \dot{Q} \left(\frac{kg}{s}\right)$$

- 체적유량 **Volume flow rate**

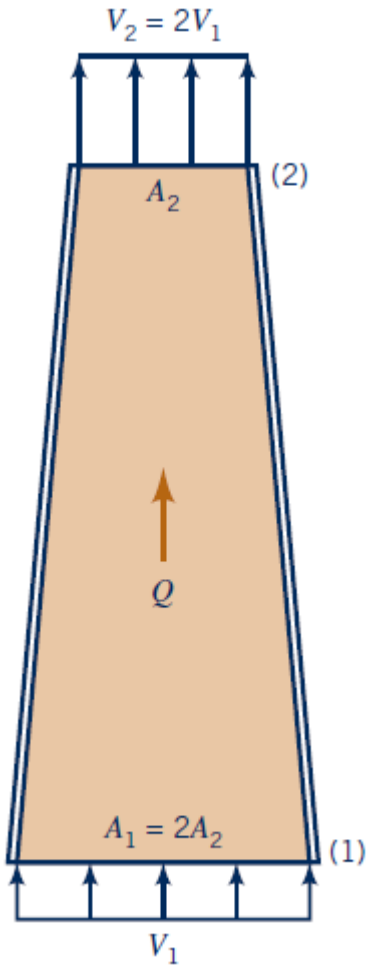
$$\dot{Q} = VA \left(\frac{m^3}{s}\right)$$

- 연속방정식 **Continuity equa**

$$\dot{m}_1 = \dot{m}_2$$

$$\therefore \rho \dot{Q}_1 = \rho \dot{Q}_2$$

$$\therefore \rho A_1 V_1 = \rho A_2 V_2$$



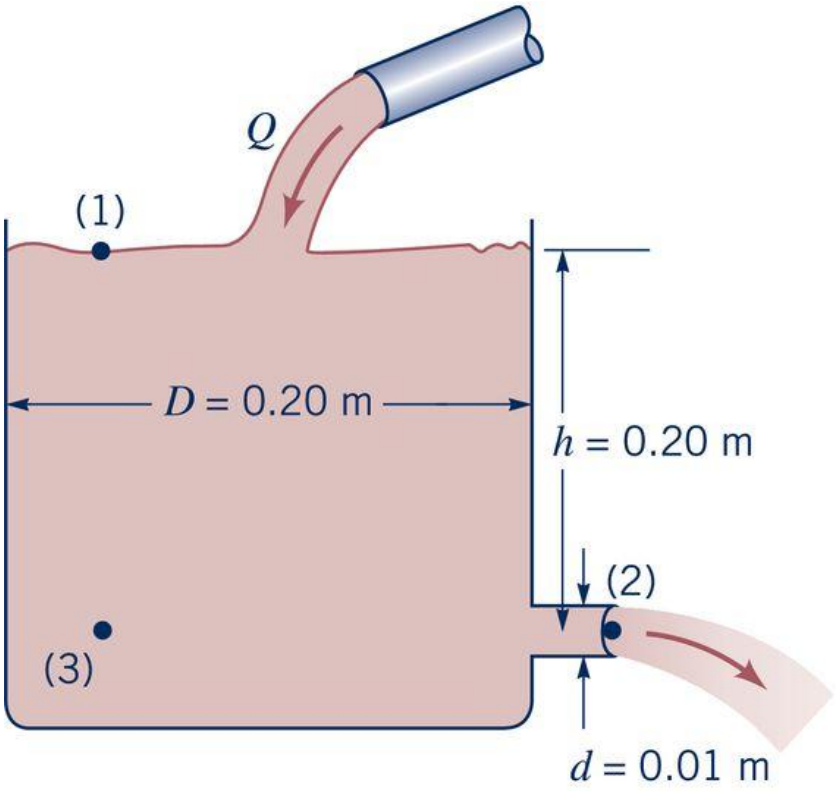
4.6 베르누이 방정식 응용 Use of the Bernoulli Equation

Ex4.7) A stream of refreshing beverage of diameter d=0.01(m) flows steadily from the cooler of diameter 0.2(m) as shown below. Determine the flow rate Q from the bottle into the cooler if the depth of beverage in the cooler is to remain constant at h=0.2(m).

Ex4.7) 직경 0.01(m)인 청량음료 물줄기가 직경 0.2(m)인 쿨러로부터 일정하게 흘러나온다. 만약 쿨러 내 음료의 수위가 0.2(m)로 일정하게 유지된다면 병에서 쿨러로 공급해야 하는 유량은 얼마인가?



(a)



(b)

4.6 베르누이 방정식 응용 Use of the Bernoulli Equation

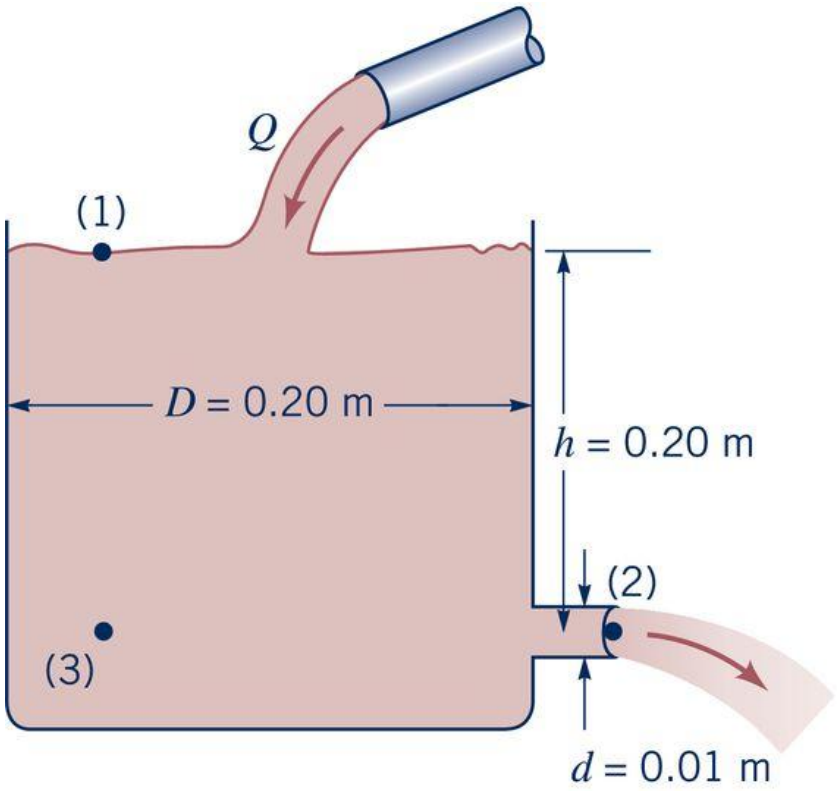
- 질량 유량 **mass flow rate**

$$\dot{m}_1 = \dot{m}_2$$

$$\therefore \rho \dot{Q}_1 = \rho \dot{Q}_2$$

$$\therefore \dot{Q}_1 = \dot{Q}_2 \Rightarrow A_1 V_1 = A_2 V_2$$

$$\therefore V_1 = \left(\frac{d}{D}\right)^2 V_2$$



(b)

4.6 베르누이 방정식 응용 Use of the Bernoulli Equation

- 점 (1)과 (2) 사이에 베르누이 방정식을 적용하면 **Bernoulli equation between (1) and (2)**

$$p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_2 + \frac{\rho V_2^2}{2} + \gamma z_2$$

$$p_1 = p_2 \approx p_{atm}, \qquad z_1 = h \ \ and \ z_2 = 0$$

$$\therefore \frac{\rho V_2^2}{2} = \frac{\rho V_1^2}{2} + \gamma h \Rightarrow \frac{V_2^2}{2} = \frac{V_1^2}{2} + gh = \frac{\left[\left(\frac{d}{D}\right)^2 V_2\right]^2}{2} + gh = \frac{1}{2}\left(\frac{d}{D}\right)^4 V_2^2 + gh$$

$$\therefore V_2 = \sqrt{\frac{2gh}{1-\left(\frac{d}{D}\right)^4}} = \sqrt{\frac{2X9.8\left(\frac{m}{s^2}\right)0.2(m)}{1-\left(\frac{0.01}{0.2}\right)^4}} = 1.98\left(\frac{m}{s}\right)$$

- 체적 유량 **Volume flow rate**

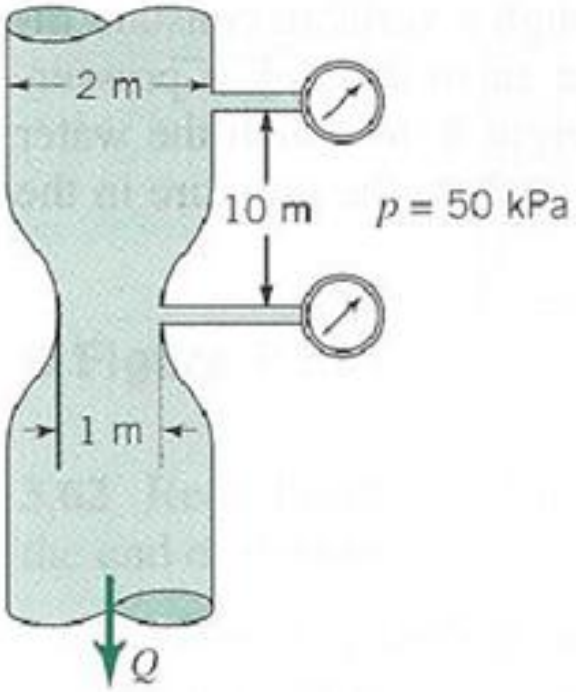
$$\dot{Q_2} = A_2 V_2 = \frac{\pi\left(0.01(m)\right)^2}{4} 1.98(\frac{m}{s}) = 1.56X10^{-4}(\frac{m^3}{s})$$



베르누이 방정식 응용 Use of the Bernoulli Equation

4.45) Water (assumed inviscid and incompressible) flows steadily in the vertical area shown in the figure. Determine the flow rate if the pressure in each of the gages reads 50(kPa).

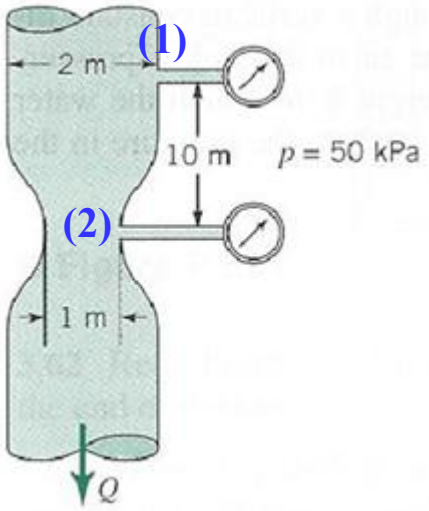
4.45) 아래 그림의 수직 단면을 물(비점성, 비압축성으로 가정)이 연속적으로 흐른다. 각 게이지 압력이 50(kPa)일 경우 유량을 구하라.





베르누이 방정식 응용 Use of the Bernoulli Equation

- 점 (1)과 (2) 사이에 베르누이 방정식 적용하면 **Bernoulli equation between (1) and (2)**



$$p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_2 + \frac{\rho V_2^2}{2} + \gamma z_2$$

$$p_1 = p_2 = 50(kPa), \qquad A_1 V_1 = A_2 V_2 \Rightarrow \frac{\pi d_1^2 V_1}{4} = \frac{\pi d_2^2 V_2}{4}$$

$$\therefore \frac{\rho(V_2^2 - V_1^2)}{2} = \gamma(z_1 - z_2)$$

$$\therefore \frac{\rho V_2^2 (1 - \frac{V_1^2}{V_2^2})}{2} = \frac{\rho V_2^2 (1 - \left(\frac{d_2}{d_1}\right)^4)}{2} = \gamma(z_1 - z_2)$$

$$\therefore V_2 = \sqrt{\frac{2g(z_1 - z_2)}{1 - \left(\frac{d}{D}\right)^4}} = \sqrt{\frac{2X9.8\left(\frac{m}{s^2}\right)10(m)}{1 - \left(\frac{1}{2}\right)^4}} = 14.5\left(\frac{m}{s}\right)$$

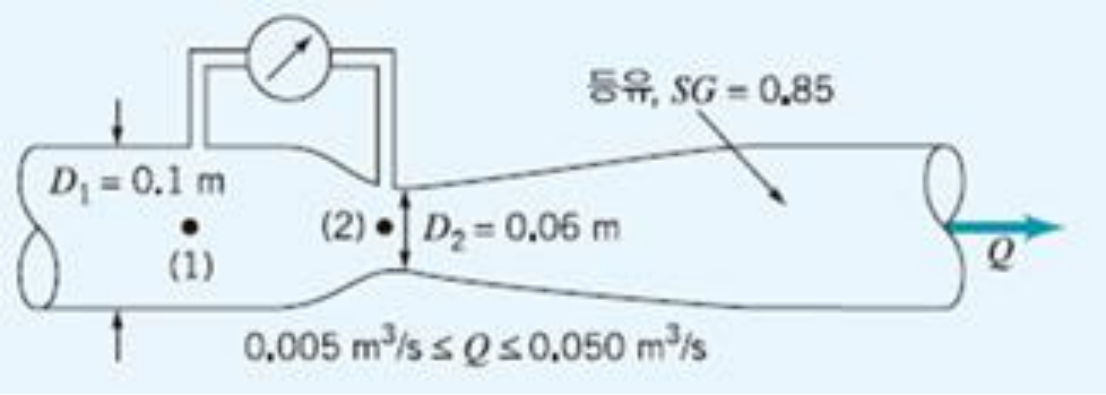
- **체적유량 volume flow rate**

$$\dot{Q_2} = A_2 V_2 = \frac{\pi\left(1(m)\right)^2}{4} 14.5(\frac{m}{s}) = 11.4(\frac{m^3}{s})$$

Quiz #1 : 베르누이 방정식 응용 Use of the Bernoulli Equation

Q1) Kerosene (SG=0.85) flows through the Venturi meter shown in the figure with flow rates between 0.005 and 0.05(m^3/s). Determine the range in pressure difference, p_1-p_2, needs to measure these flow rates.

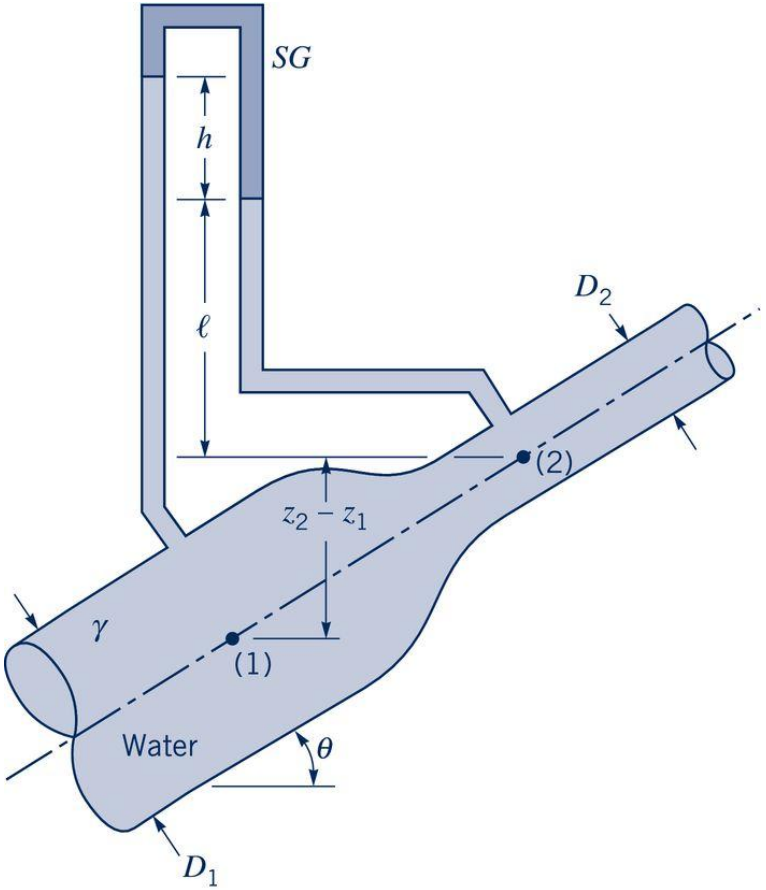
Q1) 벤튜리 속으로 등유(SG=0.85)가 유량 0.005~0.05(m^3/s)로 흐른다. 이 유량을 측정하기 위한 압력 차이 p_1-p_2 범위를 구하라. (10점)



4.6 베르누이 방정식 응용 Use of the Bernoulli Equation

Ex 4.9) Water flows through a pipe reducer as shown below. The static pressures at (1) and (2) are measured by the inverted U-tube manometer containing oil of specific gravity, SG, less than one. Determine the manometer reading.

Ex4.9) 물이 관경 축소부를 흐른다. 점(1)과 (2)에서의 압력을 역 U튜브 액주계를 사용하여 측정한다. 액주계 눈금을 구하라



4.6 베르누이 방정식 응용 Use of the Bernoulli Equation

- 점 (1)과 (2) 사이에 베르누이 방정식 **Bernoulli equation between (1) and (2)**

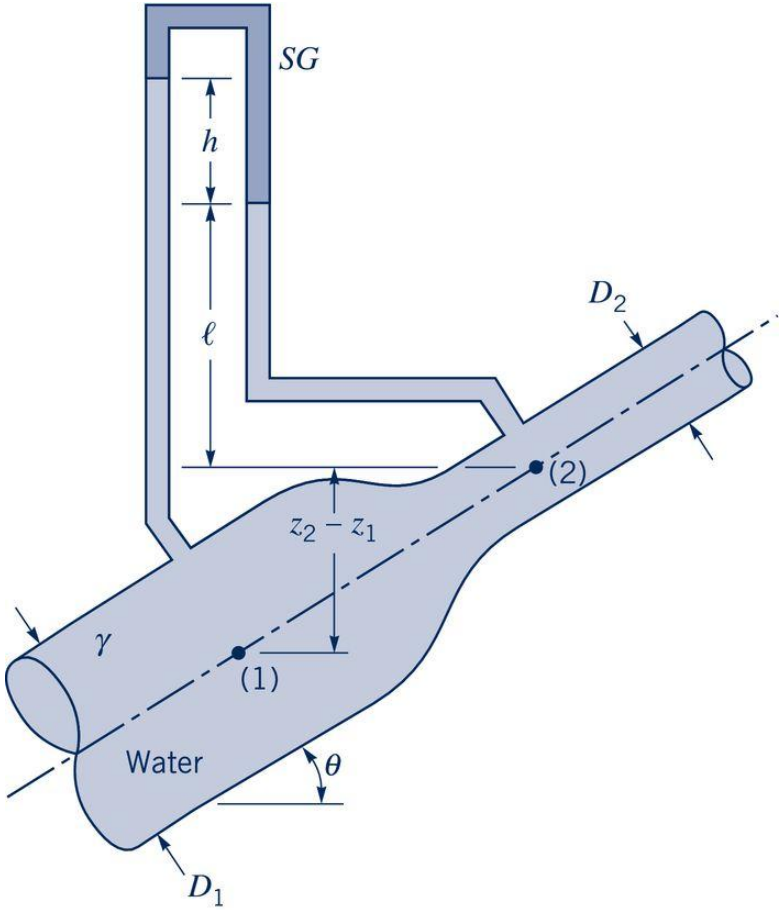
$$p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_2 + \frac{\rho V_2^2}{2} + \gamma z_2$$

$$\therefore p_1 - p_2 = \frac{\rho V_2^2}{2} - \frac{\rho V_1^2}{2} + \gamma(z_2 - z_1)$$

- 연속방정식 **Continuity equation**

$$\dot{Q} = A_1 V_1 = A_2 V_2$$

$$\therefore V_1 = \frac{\dot{Q}}{A_1} \quad and \quad V_2 = \frac{\dot{Q}}{A_2}$$



4.6 베르누이 방정식 응용 Use of the Bernoulli Equation

- 액주계 눈금 **Manometer reading**

$$p_1 - \gamma(z_2 - z_1 + l + h) + \gamma_{gage}h + \gamma l = p_2$$

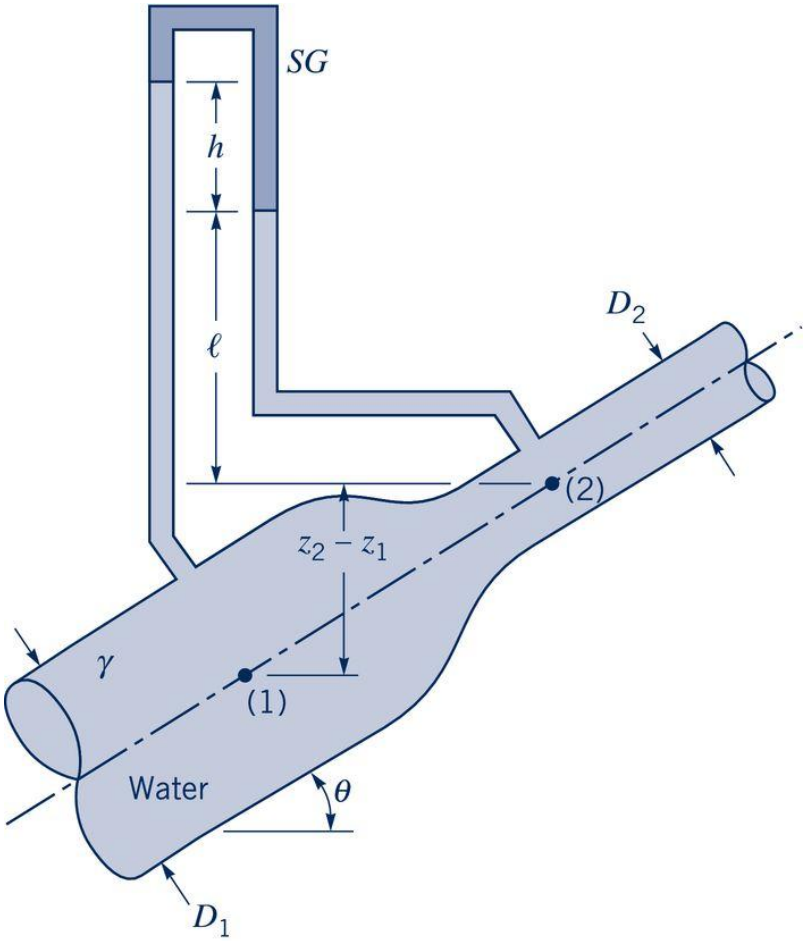
$$\therefore p_1 - p_2 = \gamma h - \gamma_{gage}h + \gamma(z_2 - z_1)$$

$$\therefore h = \frac{p_1 - p_2}{\gamma - \gamma_{gage}} - \frac{(z_2 - z_1)}{\gamma - \gamma_{gage}}$$

$$since\ p_1 - p_2 = \frac{\rho V_2^2}{2} - \frac{\rho V_1^2}{2} + \gamma(z_2 - z_1)$$

$$\therefore h = \frac{\rho(V_2^2 - V_1^2)}{2(\gamma - \gamma_{gage})}$$

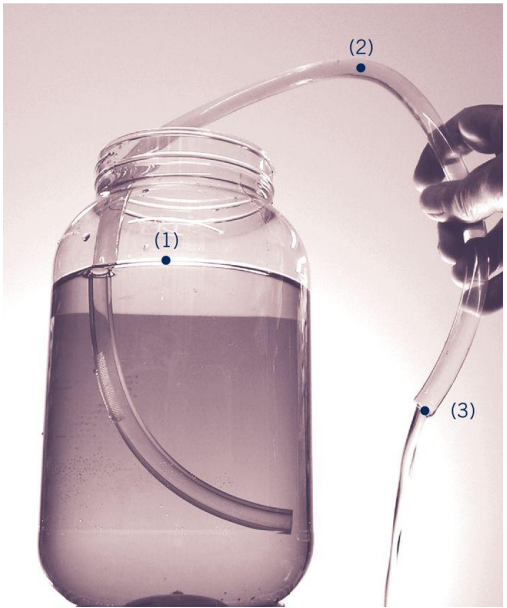
$$= \frac{\rho\left(\left(\frac{\dot{Q}}{A_2}\right)^2 - \left(\frac{\dot{Q}}{A_1}\right)^2\right)}{2\gamma(1 - SG)} = \left(\frac{\dot{Q}}{A_2}\right)^2 \frac{1 - \left(\frac{A_2}{A_1}\right)^2)}{2g(1 - SG)}$$

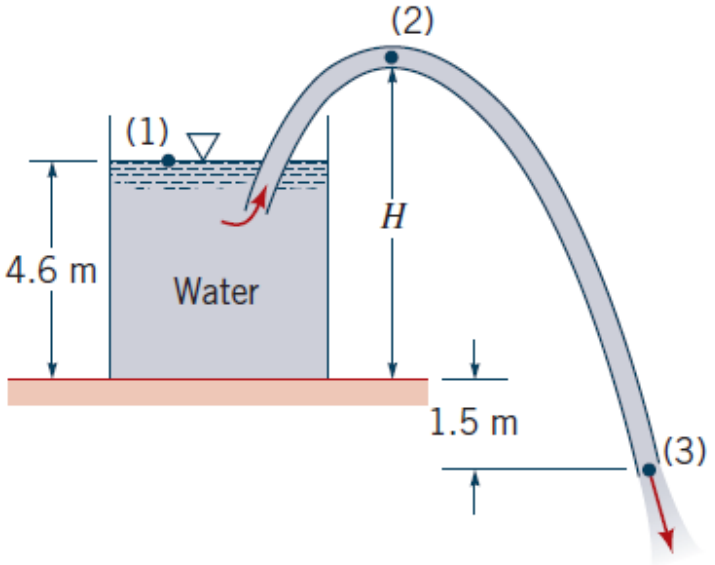


베르누이 방정식 응용 Use of the Bernoulli Equation

Ex4.10) A liquid can be siphoned from a container as shown below, provided the end of the tube, point (3), is below the free surface in the container, point (1), and the maximum elevation of the tube, point (2), is not too great. Consider water at 15(°C) being siphoned from a large tank through a constant diameter hose as shown in the figure. The end of the siphon is 1.5(m) below the bottom of the tank, and the atmospheric pressure is 101.3(kPa). Determine the maximum height of the fill H, over which water can be siphoned without cavitation occurring.

Ex4.10) 호스의 제일 높은 점 (2)가 너무 높지 않고, 호스 끝인 점 (3)이 탱크 수위 점(1)보다 낮은 곳에 위치하는 경우 용기로부터 물이 흐를 수 있다(사이폰 될 수 있다). 사이폰의 끝이 탱크 바닥보다 1.5(m)아래에 놓이고 대기압이 101.3(kPa)일 경우 캐비테이션(공동현상)이 발생하지 않는 최대 높이 H를 구하라





베르누이 방정식 응용 Use of the Bernoulli Equation

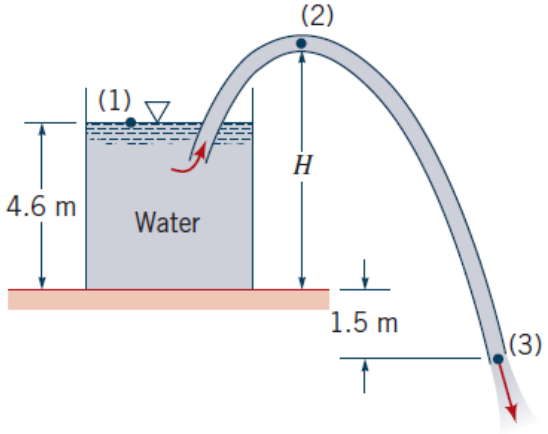
- 점 (1)과 (2) 사이에 베르누이 방정식 Bernoulli equation between (1) and (2)

$$p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_2 + \frac{\rho V_2^2}{2} + \gamma z_2$$

$$p_1 = p_{atm} \; and \; V_1 \approx 0$$

$$p_2 = p_{atm} - \frac{\rho V_2^2}{2} + \gamma(z_1 - z_2)$$

$$= p_{atm} - \frac{\rho V_2^2}{2} + \gamma(4.5 - H)$$



- 점 (1)과 (3) 사이에 베르누이 방정식Bernoulli equation between (1) and (3)

$$p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_3 + \frac{\rho V_3^2}{2} + \gamma z_3$$

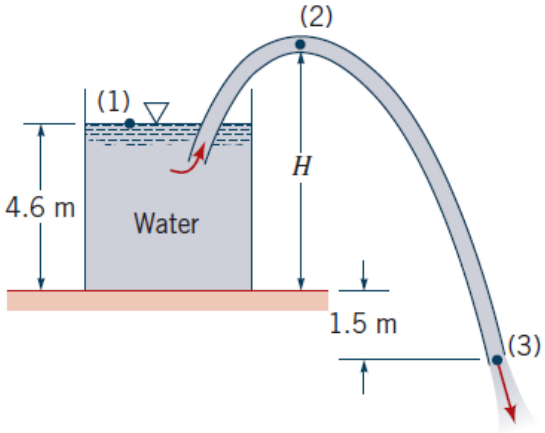
$$p_1 = p_3 = p_{atm} \; and \; V_1 \approx 0$$

$$\therefore \frac{\rho V_3^2}{2} = \gamma(z_1 - z_3)$$

$$\therefore V_3 = \sqrt{2g(z_1 - z_3)} = \sqrt{2X9.8(\frac{m}{s^2})(4.6 - (-1.5))} = 10.9\left(\frac{m}{s}\right)$$

베르누이 방정식 응용 Use of the Bernoulli Equation

- 최대높이 Maximum height H



$$p_2 = p_{atm} - \frac{\rho V_2^2}{2} + \gamma(4.6 - H)$$

$$\therefore H = 4.6 + \frac{p_{atm} - p_2}{\gamma} - \frac{V_2^2}{2g} = 4.6 + \frac{p_{atm} - p_2}{\gamma} - \frac{V_2^2}{2g}$$

$$p_2 = p_{vapor}\ and\ V_2 \approx V_3$$

$$\therefore H = 4.6 + \frac{p_{atm} - p_{vapor}}{\gamma} - \frac{V_3^2}{2g}$$

$$= 4.6(m) + \frac{(101.3 - 1.765)(kPa)X1000\left(\frac{Pa}{kPa}\right)}{1000\left(\frac{kg}{m^3}\right)9.8\left(\frac{m}{s}\right)} - \frac{\left(10.9\left(\frac{m}{s}\right)\right)^2}{2X9.8\left(\frac{m}{s^2}\right)}$$

$$= 8.6(m)$$

베르누이 방정식 응용 Use of the Bernoulli Equation

Ai

공동현상을 창의적으로 활용한 산업적 사례는?

한 가지를 선택하여 학습하기

4.6 베르누이 방정식 응용 Examples of Use of the Bernoulli Equation

4.6.3 유량 측정 Flow rate measurement : 오리피스 orifice

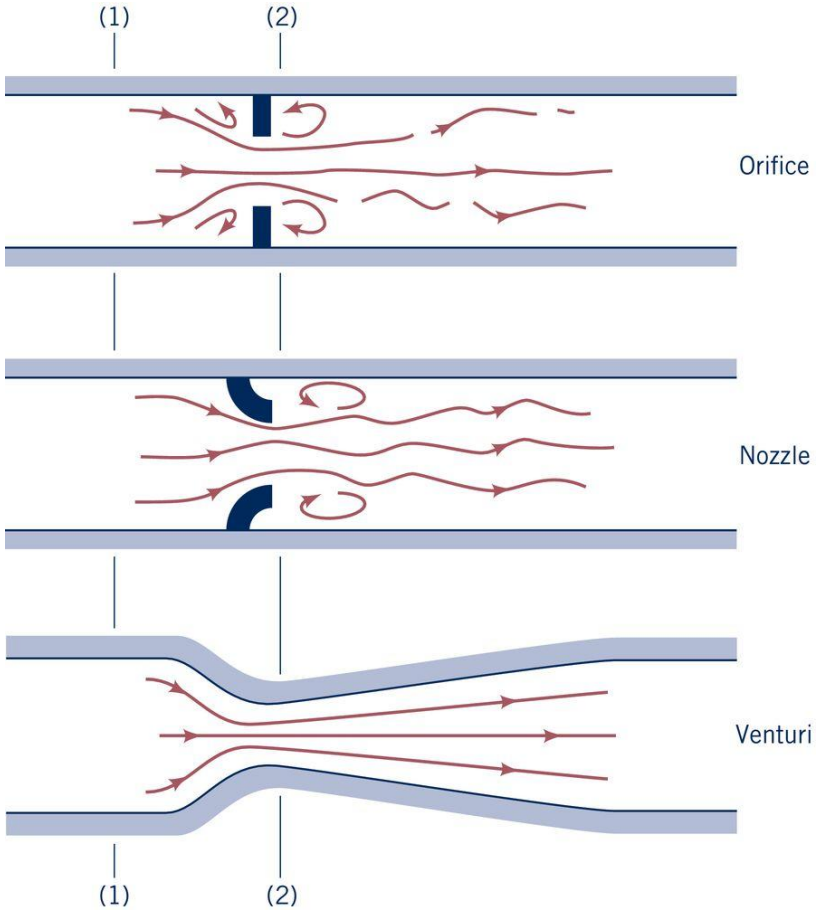
- Bernoulli equation between (1) and (2) 연속방정식 continuity equation

$$p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_2 + \frac{\rho V_2^2}{2} + \gamma z_2 \qquad \dot{Q} = A_1 V_1 = A_2 V_2 \therefore V_1 = \frac{A_2 V_2}{A_1}$$

$$\therefore \frac{\rho V_2^2}{2} - \frac{\rho V_1^2}{2} = p_1 - p_2$$

$$\therefore V_2^2 \left(1 - \frac{V_1^2}{V_2^2}\right) = V_2^2 \left(1 - \left(\frac{A_2}{A_1}\right)^2\right) = \frac{2(p_1 - p_2)}{\rho}$$

$$\therefore V_2 = \sqrt{\frac{2(p_1 - p_2)}{\rho\left(1 - \left(\frac{A_2}{A_1}\right)^2\right)}}$$



- 체적유량 Volume flow rate

$$\dot{Q} = A_2 V_2 = A_2 \sqrt{\frac{2(p_1 - p_2)}{\rho\left(1 - \left(\frac{A_2}{A_1}\right)^2\right)}}$$

4.6 베르누이 방정식 응용 Use of the Bernoulli Equation

4.6.3 Flow rate measurement : sluice gate

- Bernoulli equation between (1) and (2)

$$p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_2 + \frac{\rho V_2^2}{2} + \gamma z_2$$

$$\therefore \frac{\rho V_2^2}{2} - \frac{\rho V_1^2}{2} = \gamma(z_1 - z_2)$$

$$\therefore V_2^2\left(1 - \frac{V_1^2}{V_2^2}\right) = V_2^2\left(1 - \left(\frac{A_2}{A_1}\right)^2\right) = 2g(z_1 - z_2)$$

$$\therefore V_2 = \sqrt{\frac{2(z_1 - z_2)}{\left(1 - \left(\frac{A_2}{A_1}\right)^2\right)}}$$

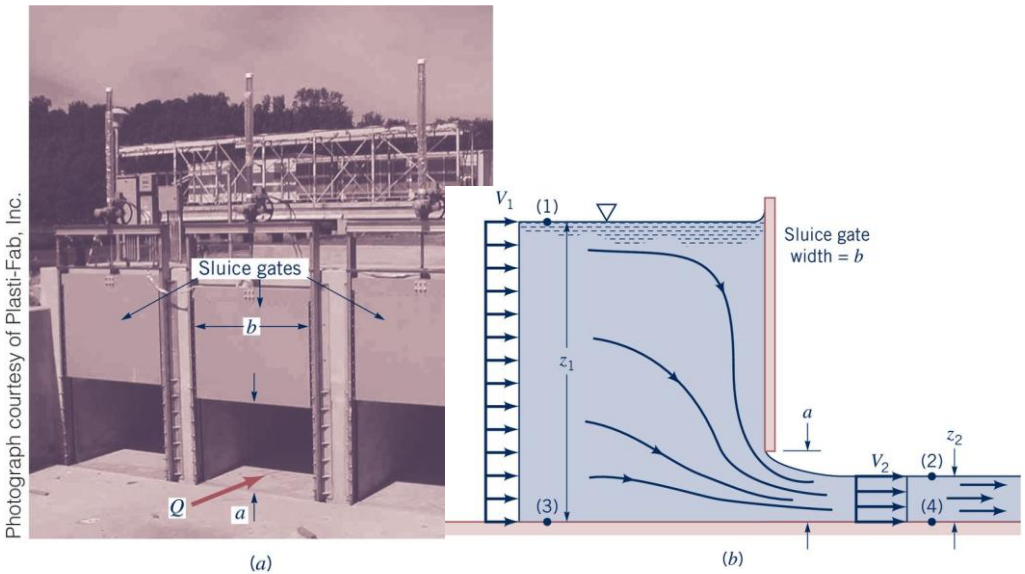
- 연속방정식 continuity equation

$$\dot{Q} = A_1 V_1 = A_2 V_2 \therefore V_1 = \frac{A_2 V_2}{A_1}$$

Volume flow rate체적 유량

$$\dot{Q} = A_2 V_2 = A_2 \sqrt{\frac{2(z_1 - z_2)}{\left(1 - \left(\frac{A_2}{A_1}\right)^2\right)}}$$

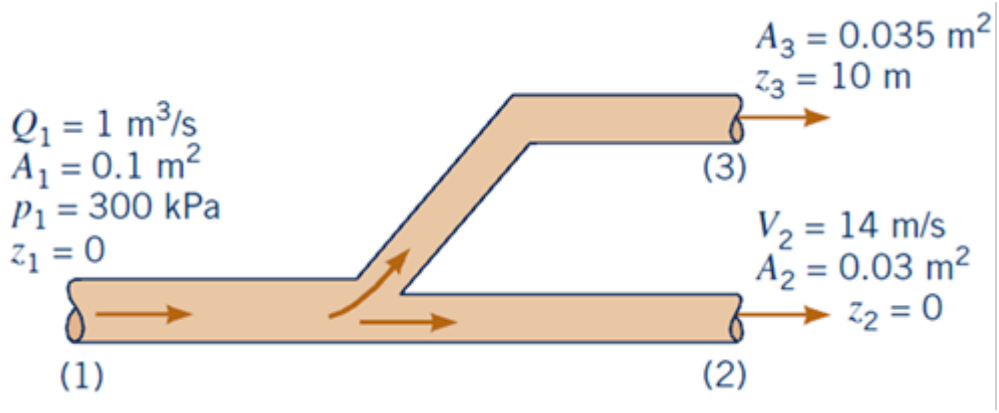
$$= z_2 b \sqrt{\frac{2(z_1 - z_2)}{\left(1 - \left(\frac{z_2}{z_1}\right)^2\right)}}$$



Quiz #2 : 베르누이 방정식 응용 Use of the Bernoulli Equation

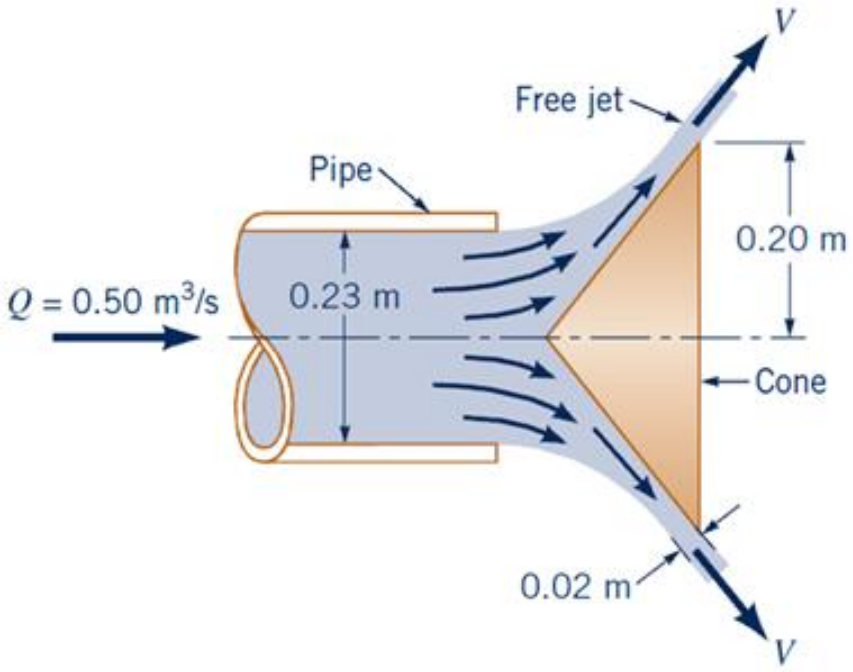
Q2) Water flows through the branching pipe shown below. If viscous effects are negligible, determine the pressure at section (2) and the pressure at section (3).

Q2) 물이 분기관을 통하여 흐른다. 점성효과를 무시할 때, 단면 (2)에서의 압력과 단면 (3)에서의 압력을 구하라. (20점)



베르누이 방정식 응용 Use of the Bernoulli Equation

- A conical plug is used to regulate the airflow from the pipe shown below. The air leaves the edge of the cone with a uniform thickness of 0.02(m). If viscous effects are negligible and the flow rate is 0.5(m^3/s), determine the pressure within the pipe.
- 파이프로부터의 기류를 원추형 플러그를 이용하여 조절한다. 공기가 콘의 가장자리를 일정한 두께 0.02(m)로 떠난다. 점성효과를 무시하고 유량이 0.5(m^3/s)인 경우, 파이프 내의 압력을 구하라.



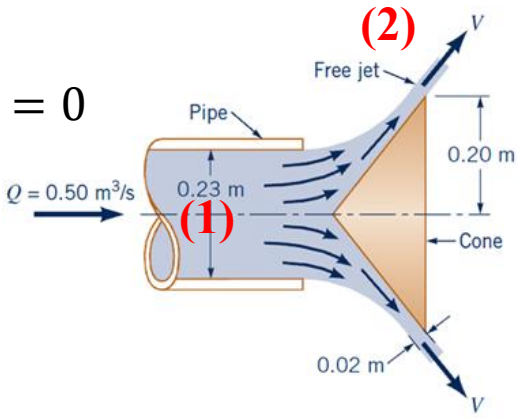
베르누이 방정식 응용 Use of the Bernoulli Equation

- 지점(1)과 (2) 사이에 베르누이 방정식 Bernoulli equation between (1) and (2)

$$p_1 + \frac{\rho V_1^2}{2} + \gamma z_1 = p_2 + \frac{\rho V_2^2}{2} + \gamma z_2$$

$$V_1 = \frac{\dot{Q}}{A_1} = \frac{0.5({}^{m^3}/_{s})}{\pi(0.23)^2/_4\ (m^2)} = 12\left(\frac{m}{s}\right), z_1 = z_2\ and\ p_2 = 0$$

$$V_2 = \frac{\dot{Q}}{A_2} = \frac{0.5({}^{m^3}/_{s})}{2\pi 0.2(m)0.02(m)} = 19.9\left(\frac{m}{s}\right)$$



$$\therefore p_1 = \frac{\rho V_2^2}{2} - \frac{\rho V_1^2}{2} = \frac{1\left({}^{kg}/_{m^3}\right)}{2}(19.9^2 - 12^2)\left(\frac{m}{s}\right)^2 = 126(Pa)$$

베르누이 방정식 응용 Use of the Bernoulli Equation

Ai

활용한 산업적 사례는?

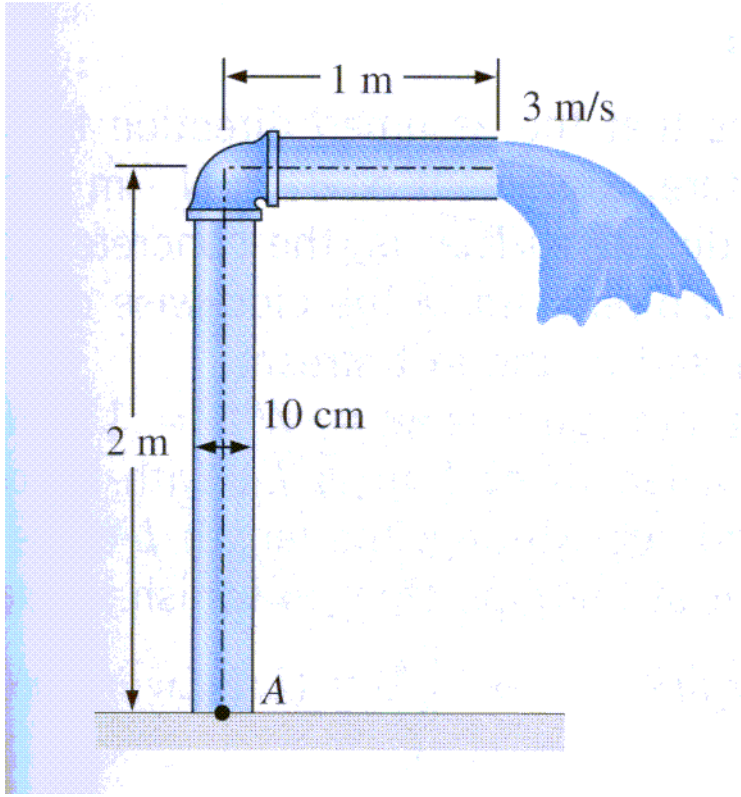
한 가지를 선택하여 학습하기

Quiz #3 : 베르누이 방정식 응용 Use of the Bernoulli Equation

Q3) 아래 그림과 같은 배관계에서 다음 물음에 답하라. 단 물의 밀도는 1000 (kg/m^3)로 가정하라. 배관계 내 유동의 점성영향을 무시하라.(20점)

(a) 체적유량 $\dot{Q}$을 구하라 (10점)

(b) A점 압력을 구하라 (10점)



4.8 베르누이 방정식 사용에 대한 제한조건
Restrictions on Use of the Bernoulli Equation

4.8.1 압축성 효과 Compressibility Effects

- **Along a streamline**

$$\frac{V^2}{2} + \int \frac{dp}{\rho} + gz = constant$$

- 유동에 대한 정보 필요 **Needs an information about the flow (process)**
- 등온 유동 **Isothermal flow : T=constant**

$$\frac{V^2}{2} + \int \frac{dp}{\rho} + gz = constant \; and \; p = \rho \cdot R \cdot T$$

$$\therefore \frac{V^2}{2} + \int \frac{dp}{p/(R \cdot T)} + gz = \frac{V^2}{2} + R \cdot T \cdot lnp + g = constant$$

$$or \;\; \frac{V_1^2}{2} + R \cdot T \cdot lnp_1 + gz_1 = \frac{V_2^2}{2} + R \cdot T \cdot lnp_2 + gz_2$$

4.8 베르누이 방정식 사용에 대한 제한조건
Restrictions on Use of the Bernoulli Equation

4.8.1 압축성 효과 Compressibility Effects

- 등 엔트로피 유동 Isentropic flow :

$$\frac{p}{\rho^k} = constant$$

$$\therefore \frac{V^2}{2} + \frac{k}{k-1}\frac{p}{\rho} + gz = constant$$

$$or \quad \frac{V_1^2}{2} + \frac{k}{k-1}\frac{p_1}{\rho_1} + gz_1 = \frac{V_2^2}{2} + \frac{k}{k-1}\frac{p_2}{\rho_2} + gz_2$$

4.8 베르누이 방정식 사용에 대한 제한조건
Restrictions on Use of the Bernoulli Equation

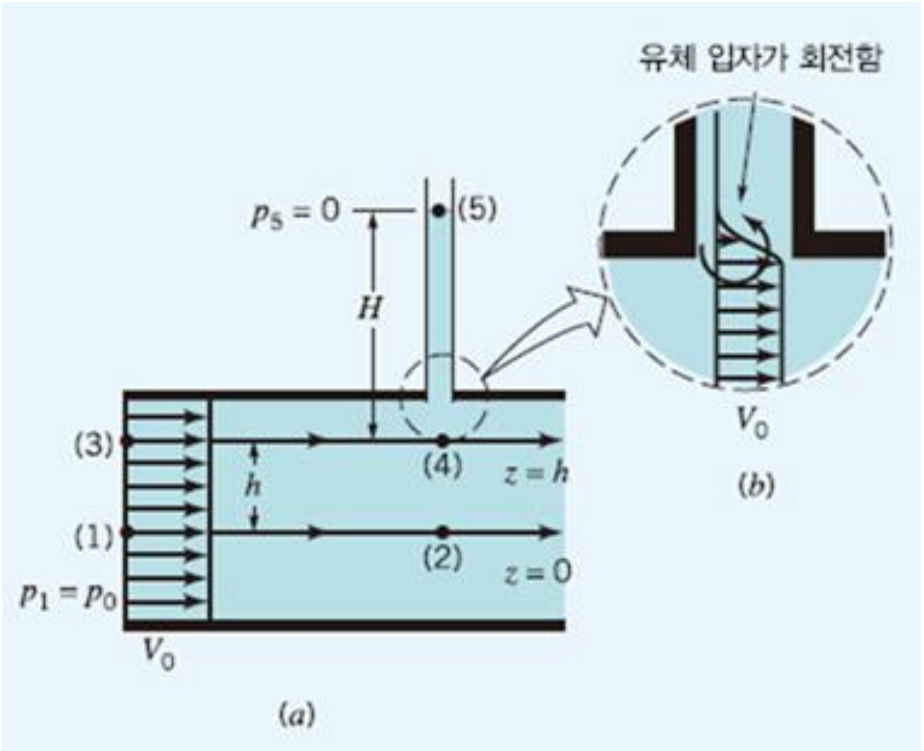
4.8.2 비정상 효과 Unsteady Effects

- **Along a streamline**

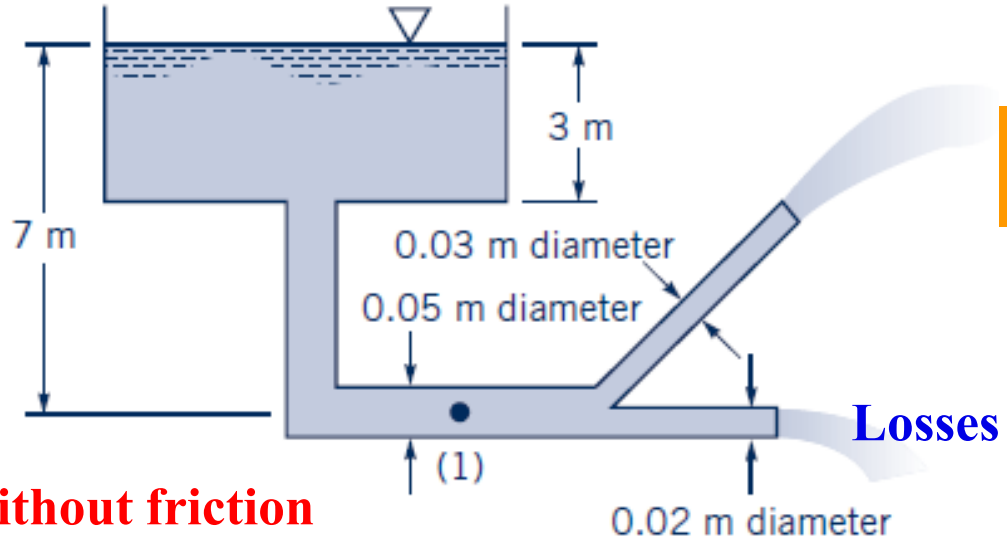
$$\rho\frac{\partial V}{\partial t}+\frac{dp}{ds}+\rho V\frac{dV}{ds}+\gamma\frac{dz}{ds}=0$$

$$\therefore \rho\frac{\partial V}{\partial t}ds+dp+\frac{\rho}{2}d(V^2)+\gamma dz=0$$

- **3.83. 회전효과 Rotational Effects**



Checklist



$$Q_{w/o\ \text{friction}} = V_{w/o\ \text{friction}} A$$

Losses : chapter 7 & 8

Without friction

$$\frac{V^2}{2} + \int \frac{dp}{\rho} + gz = constant$$

$$\frac{V^2}{2} + \frac{p}{\rho} + gz = constant: \ \ incompressible$$ 비압축성

$$or \ \ \frac{V_1^2}{2} + \frac{p_1}{\rho} + gz_1 = \frac{V_2^2}{2} + \frac{p_2}{\rho} + gz_2$$

Home work

3.5 그림 P3.5와 같이 단면적이 변하는 수직 파이프를 통하여 정상적으로 물이 흐르고 있다. 중심선을 따라서 속도는 $V = 3(1 + x)\hat{\mathbf{i}}$ m/s이고, x의 단위는 m이다. 점성효과는 무시한다. (**a**) 이 유동을 발생시키는 데 필요한 압력구배 $\partial p/\partial x$ (x의 함수로)를 구하라. (**b**) 단면 (1)에서 압력이 350 kPa일 경우 (i) (**a**)에서 구한 압력구배를 적분하여, (ii) Bernoulli방정식을 적용하여 단면 (2)에서의 압력을 각각 구하라.

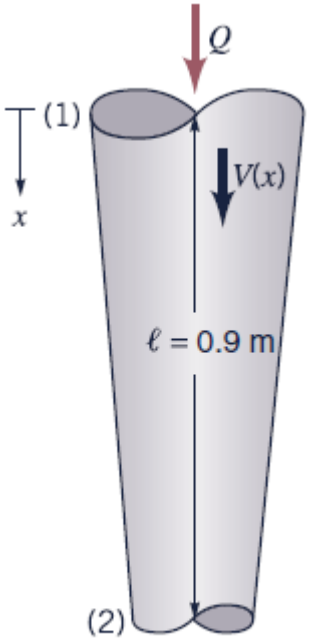


그림 P3.5

Home work

3.22 그림 P3.22와 같이 물이 큰 탱크를 통하여 흐르고 있다. 물의 깊이 h_A를 구하라.

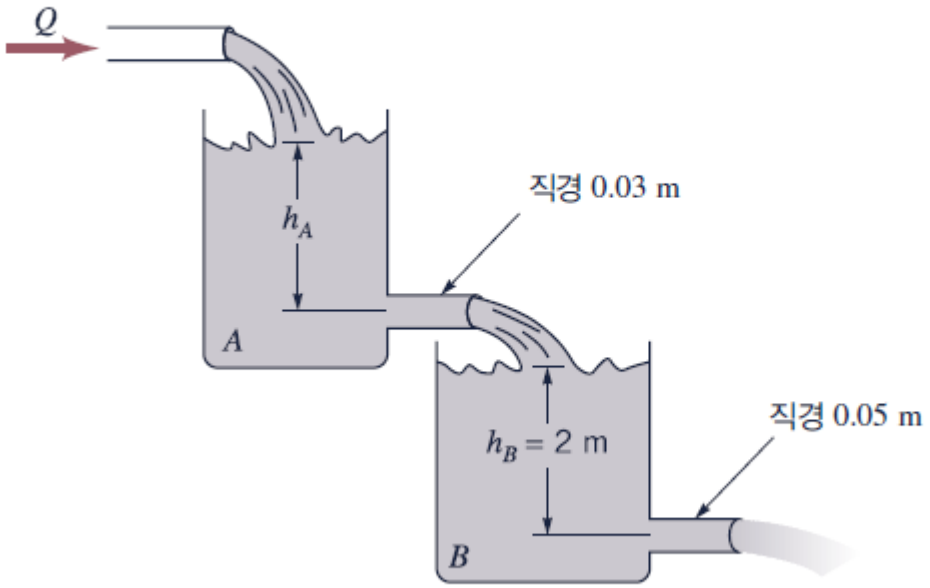


그림 P3.22